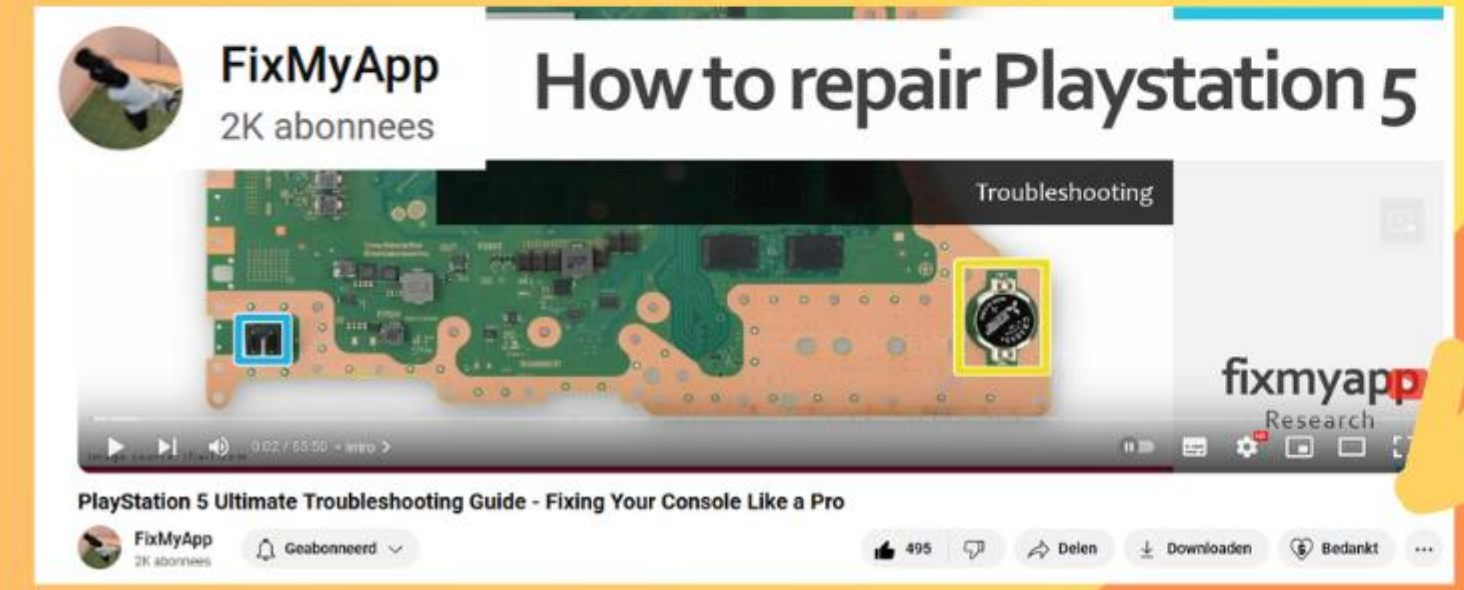
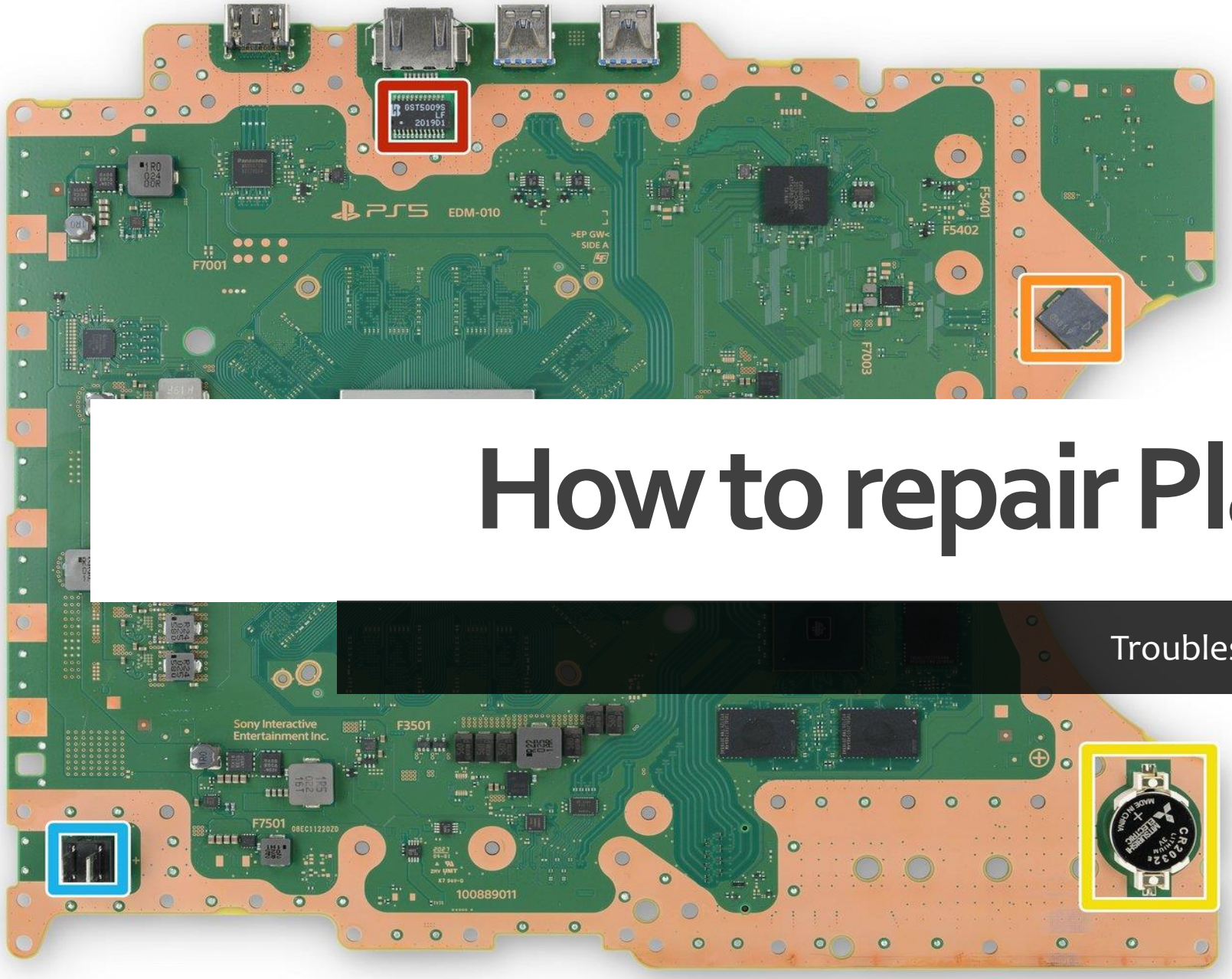
SUPPORT FIXMYAPP YOUTUBE CHANNEL WITH A

SUPER THANKS





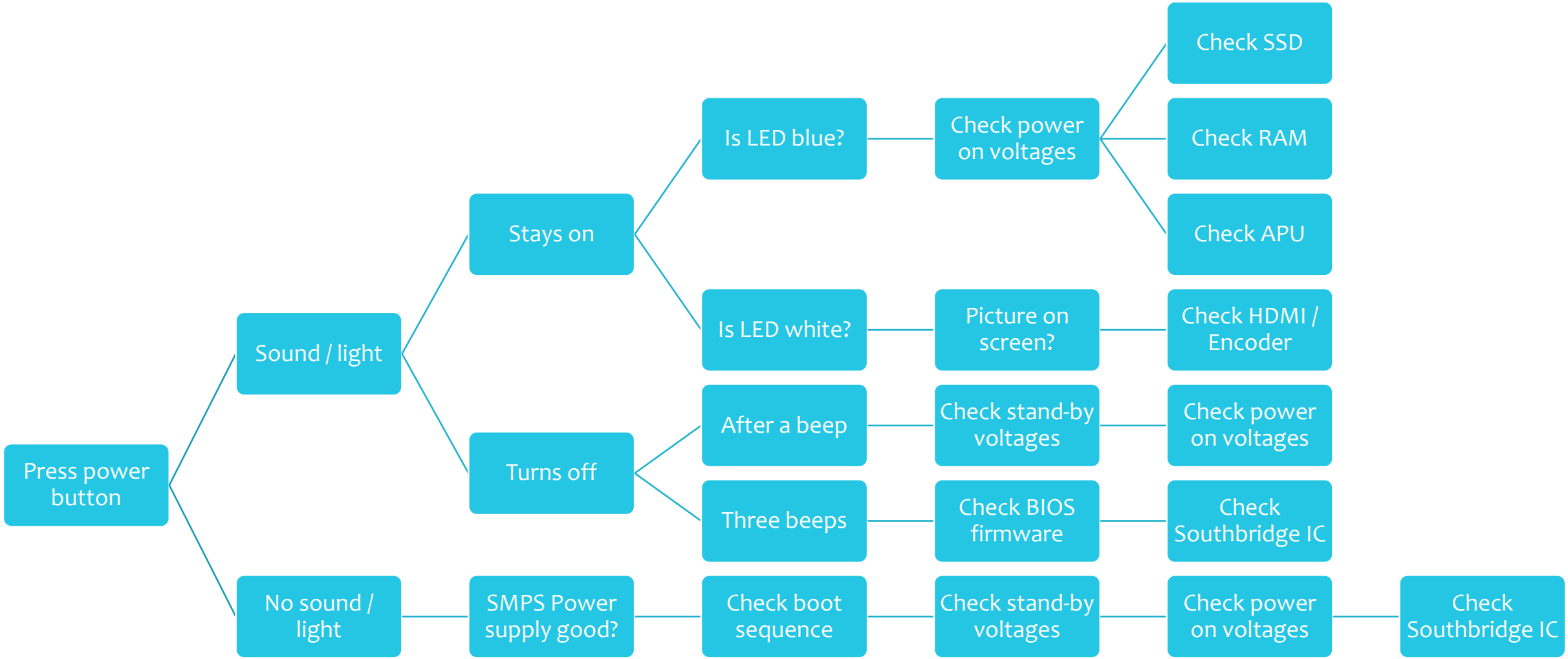
How to repair Playstation 5

Troubleshooting

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Research

Image source: ifixit.com

Flowdiagram troubleshooting



PS5 – Front architecture

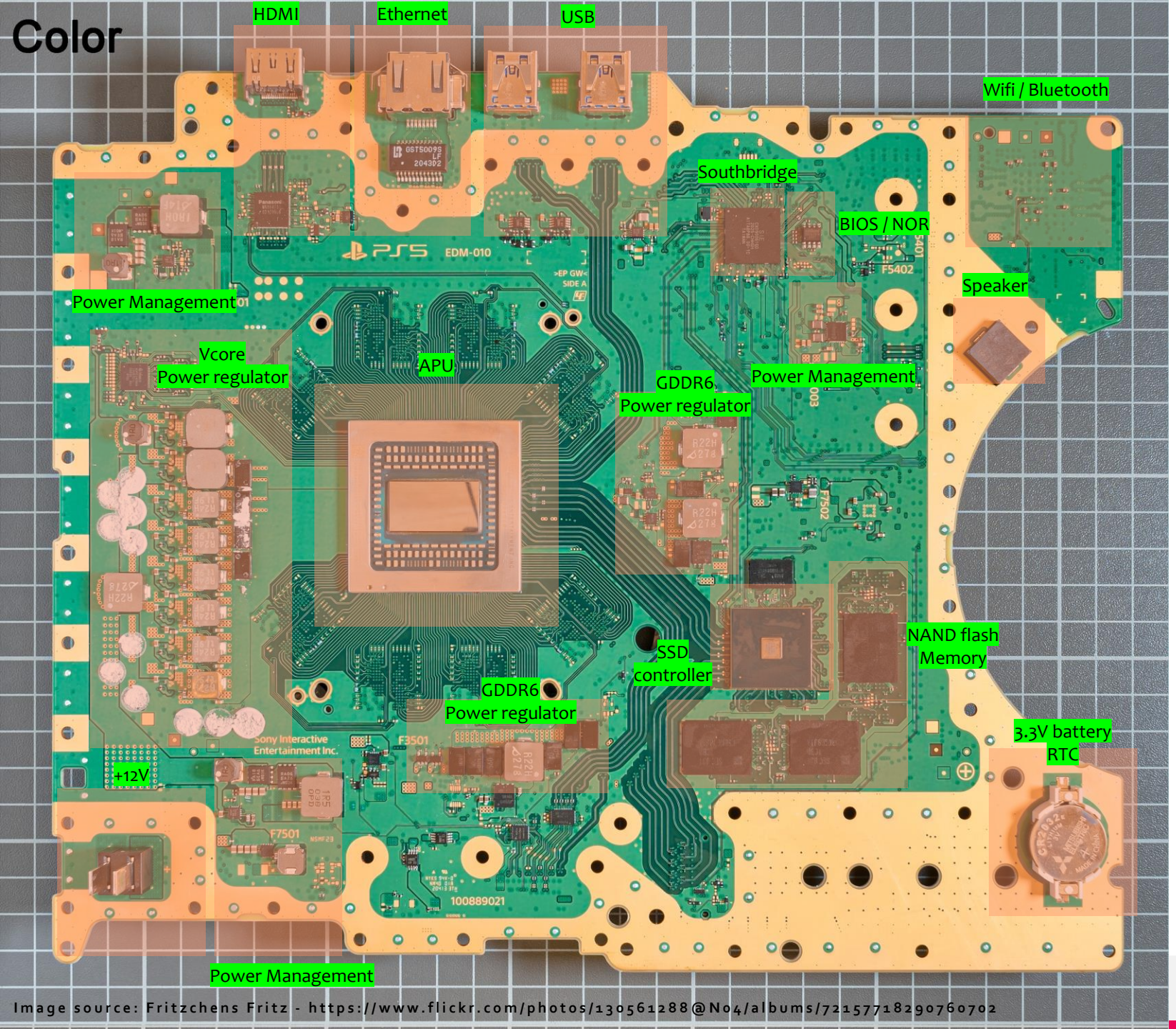


Image source: Fritzchens Fritz - https://www.flickr.com/photos/130561288@N04/albums/72157718290760702

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PS5 – Back architecture

Color

Southbridge

Wifi /Bluetooth

LED connector

FAN

Power Regulator

Power Regulator SSD

NAND flash Memory

External storage

GDDR6

Power Regulator

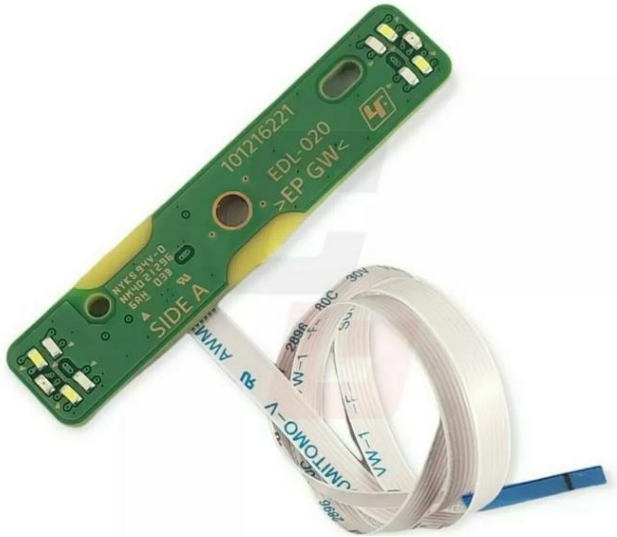
Power Regulator

APU

Vcore Power regulator

GDDR6

Power button



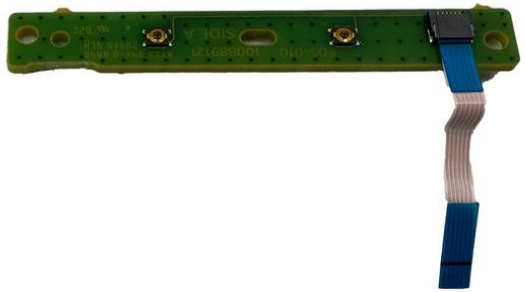
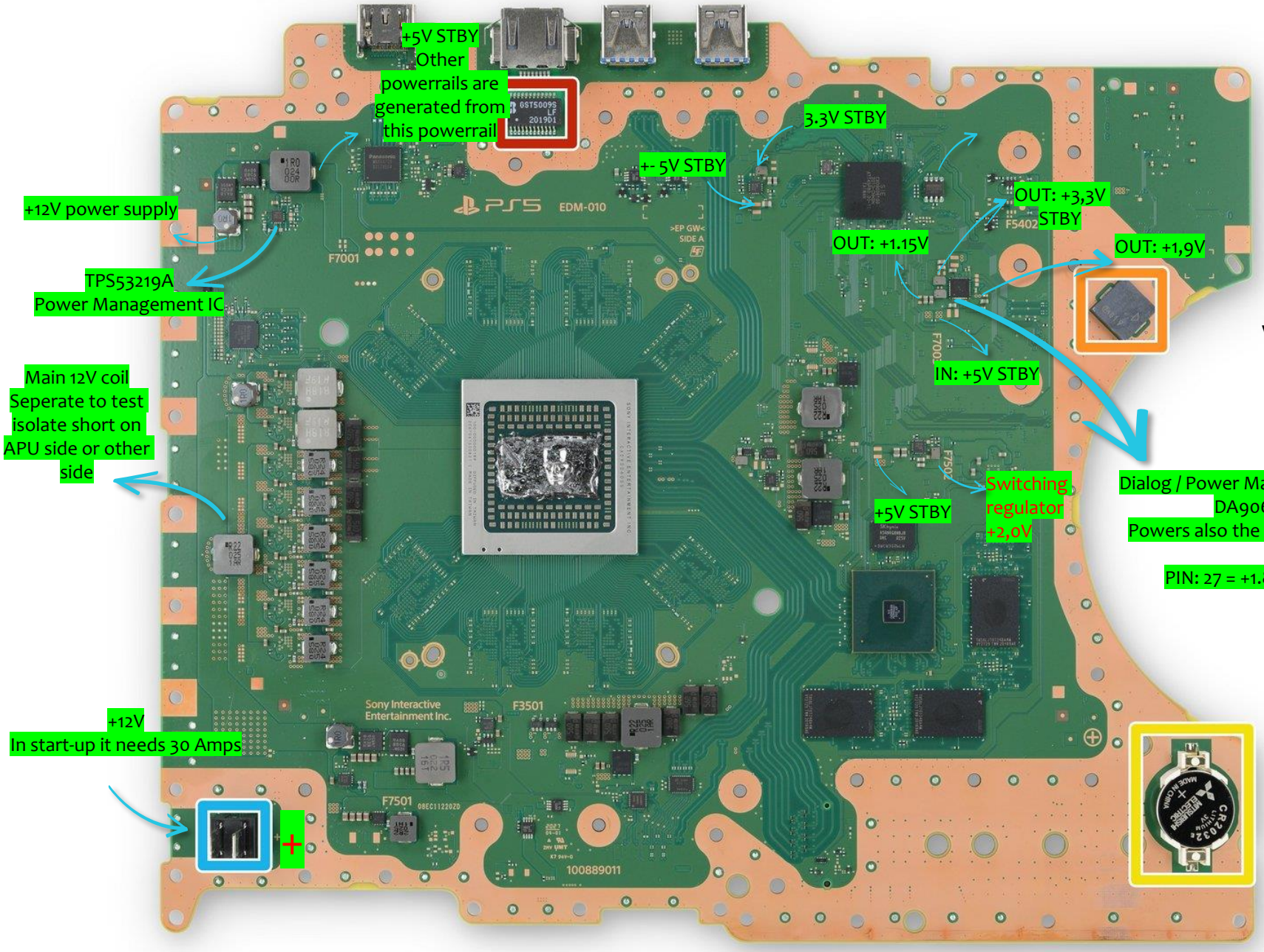


Image source: Fritzchens Fritz - https://www.flickr.com/photos/130561288@N04/albums/72157718290760702

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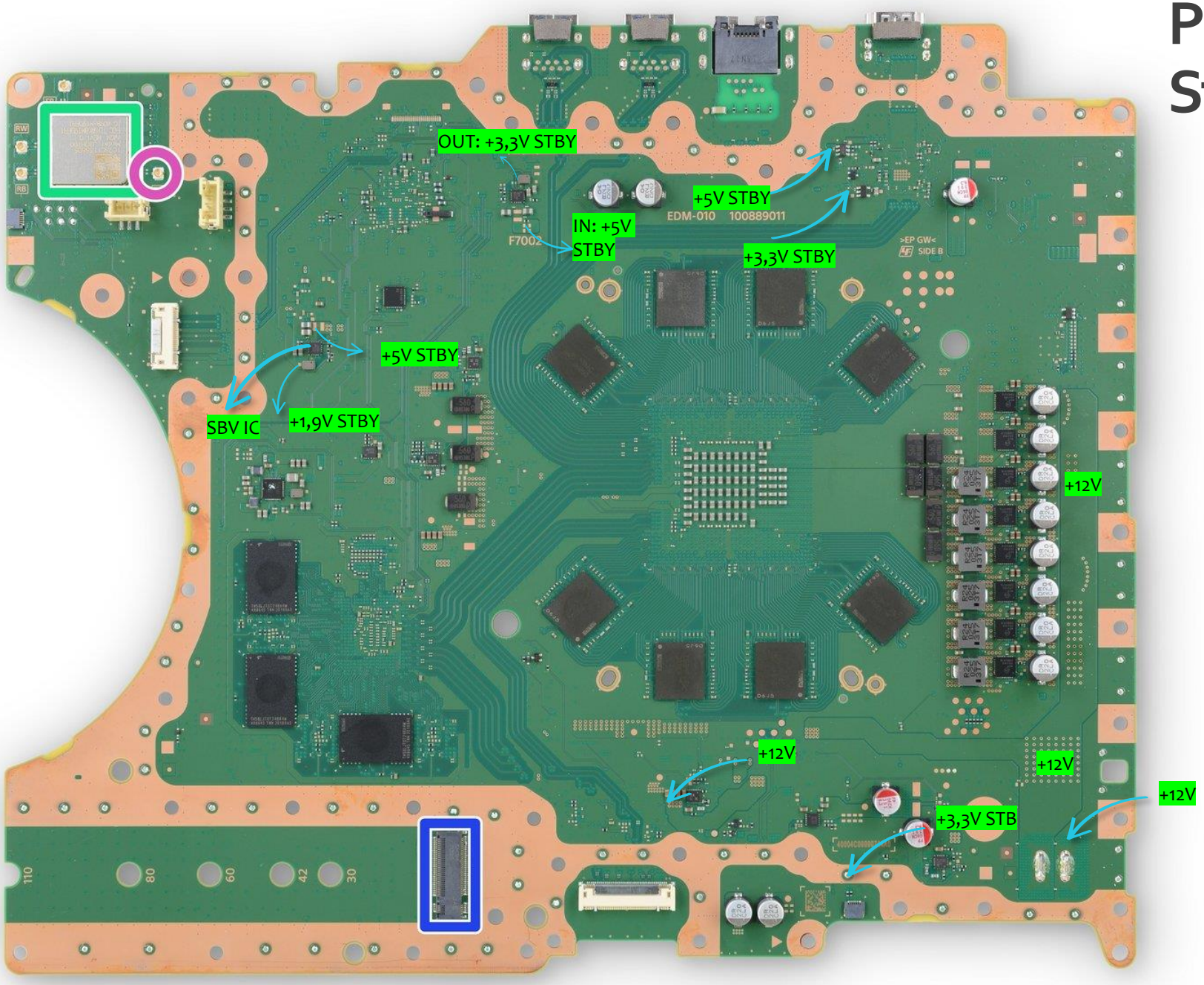
PS5 – Front Standby Voltages



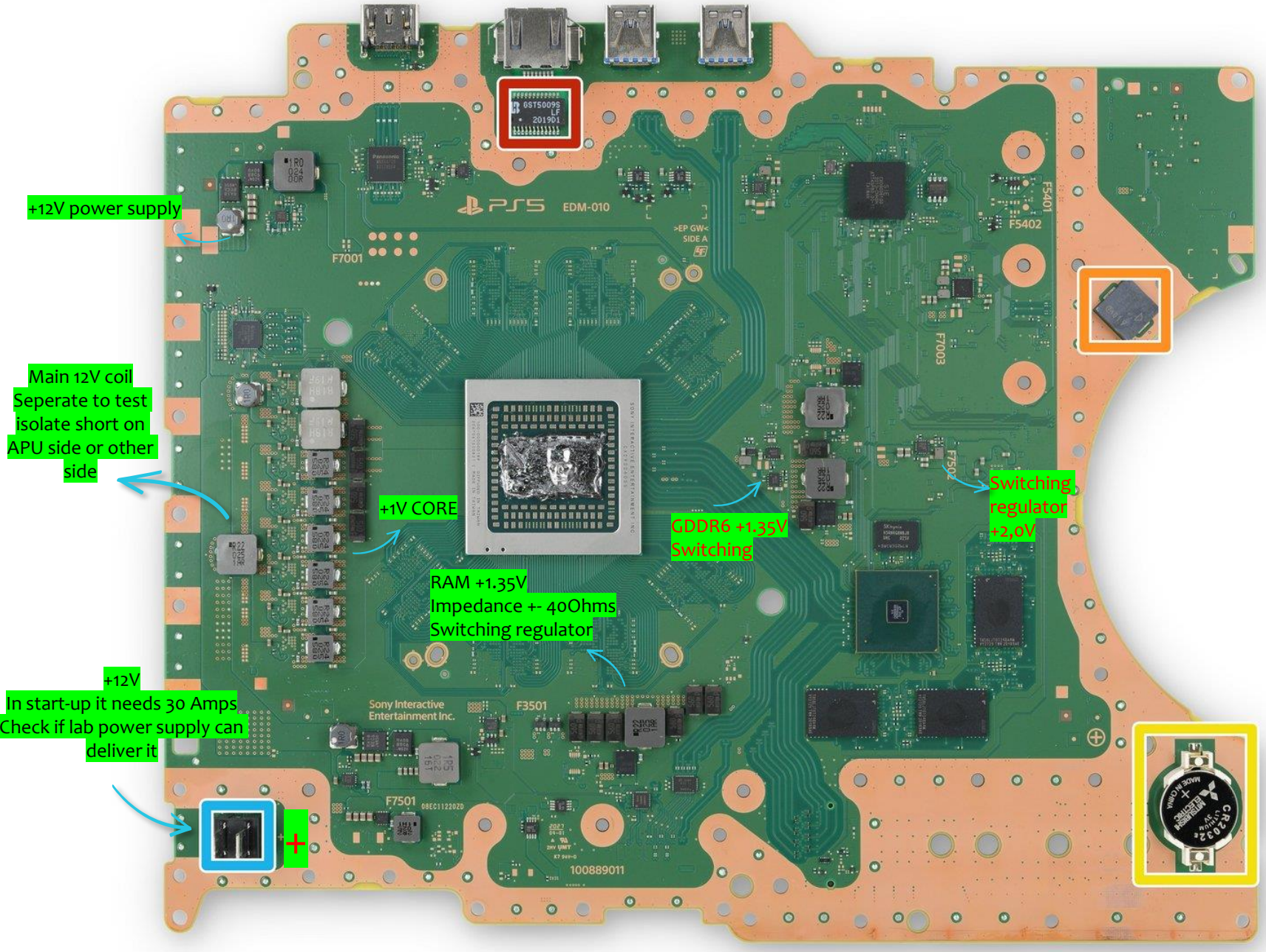
WIFI, southbridge and SSD controller, BIOS needs power in standby mode.

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PS5 – Back Standby Voltages

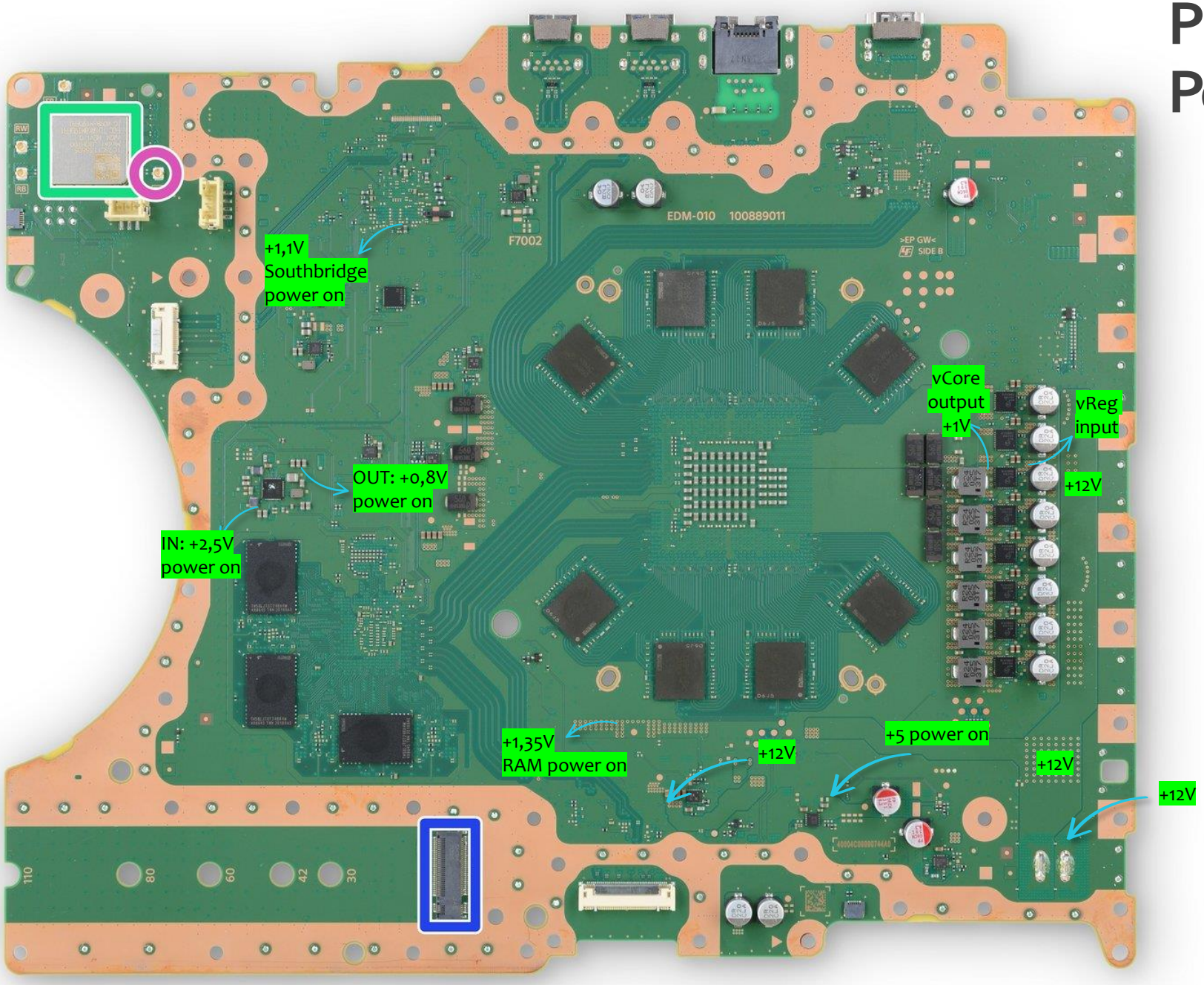


PS5 – Front Power On Voltages



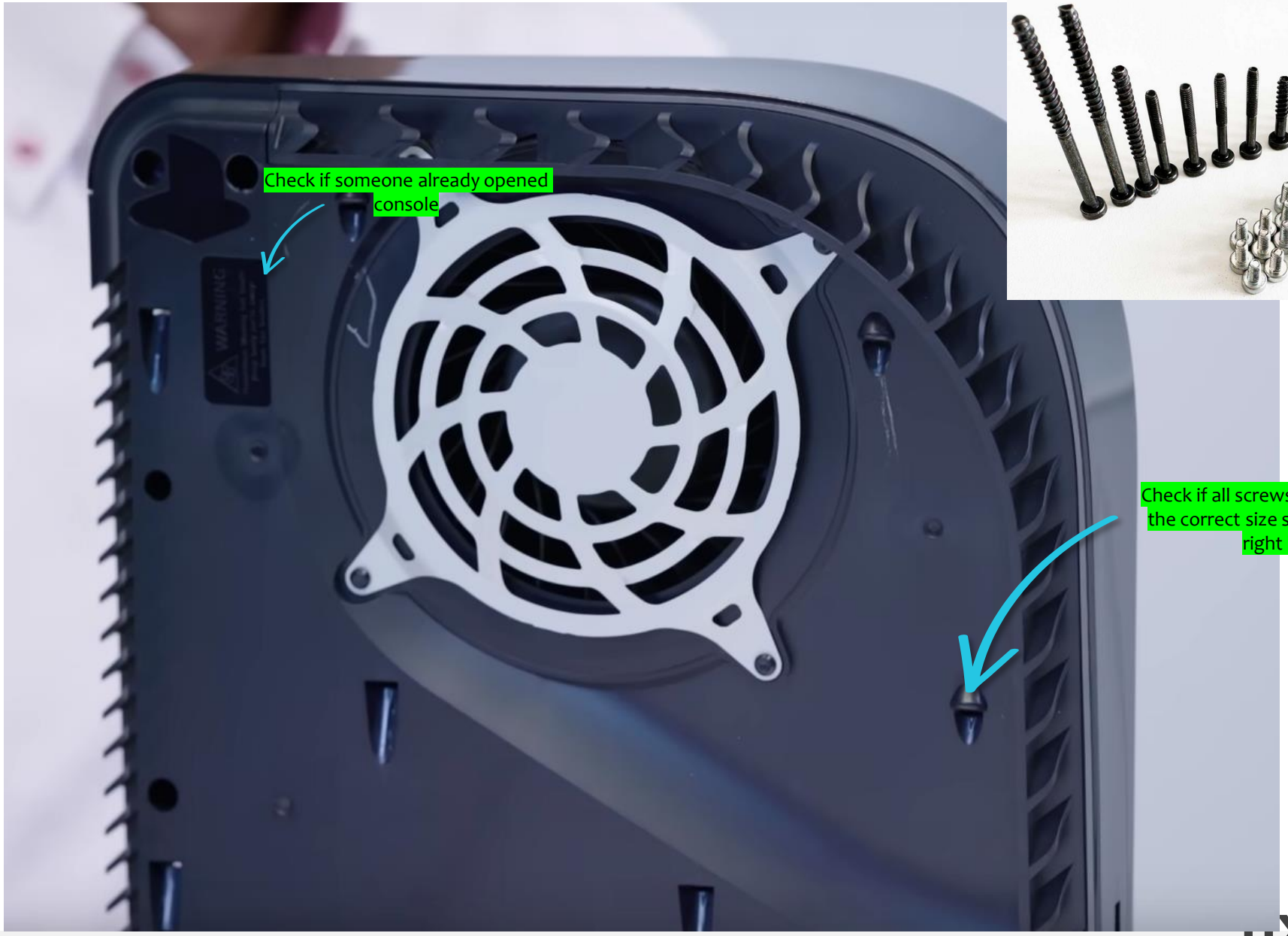
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PS5 – Back
Power On Voltages

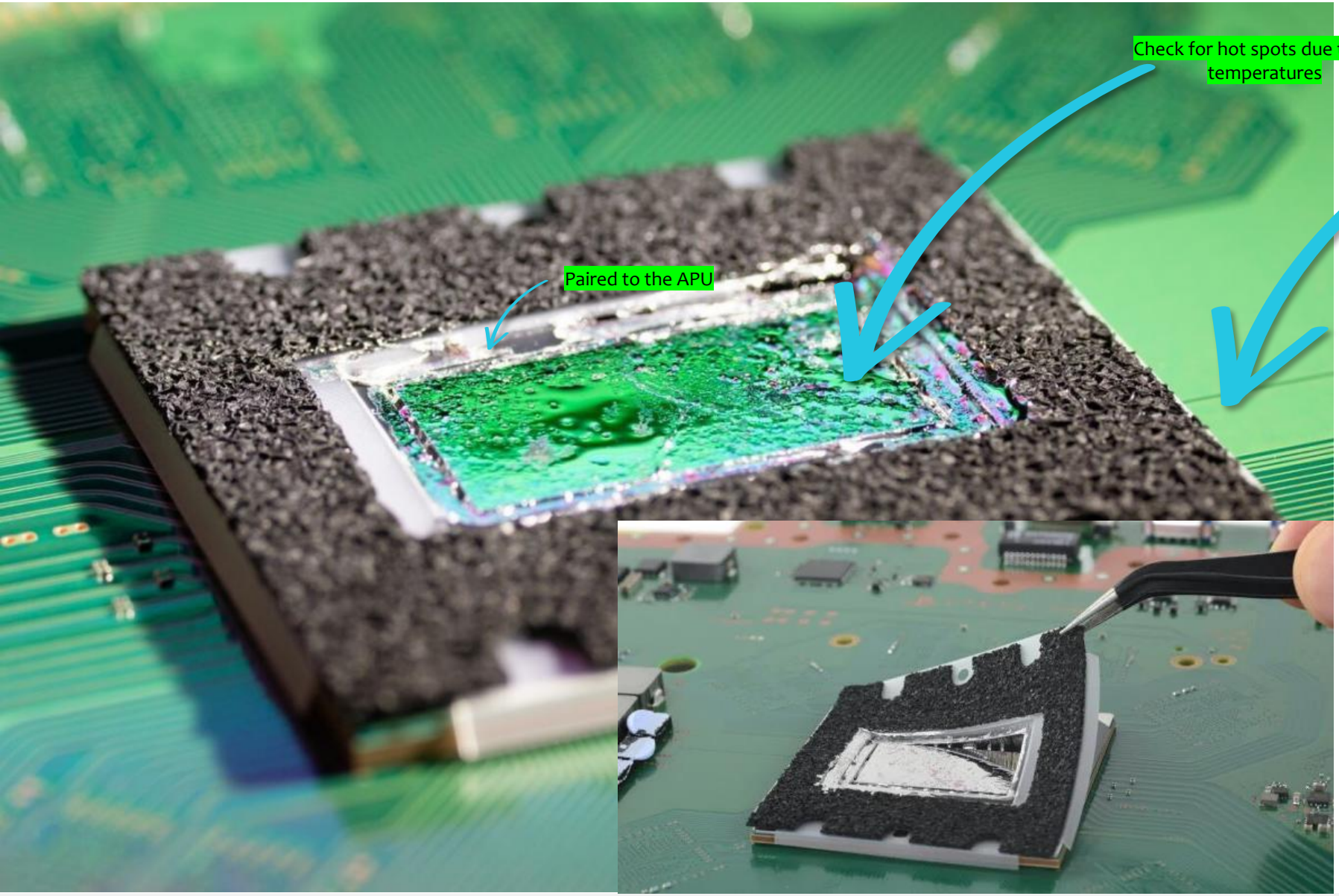


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Research

Warrenty seal and screws

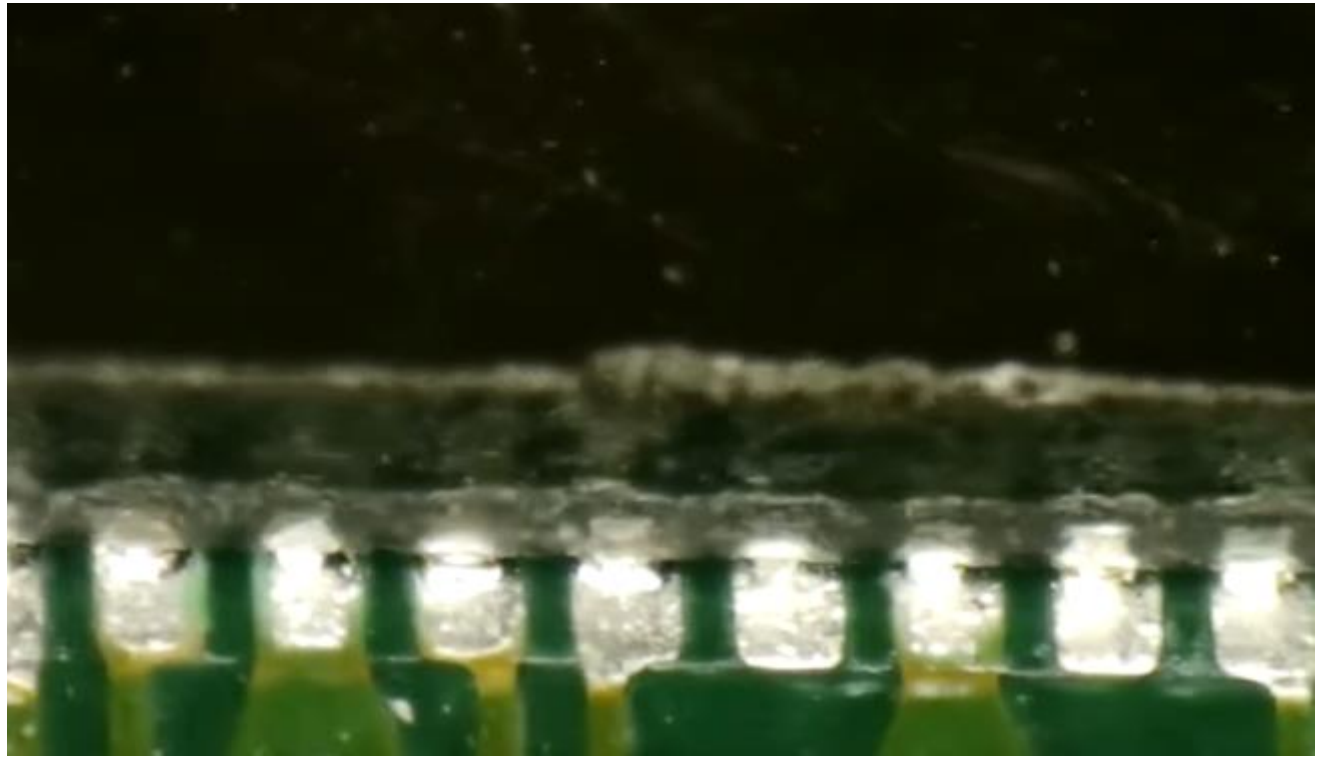


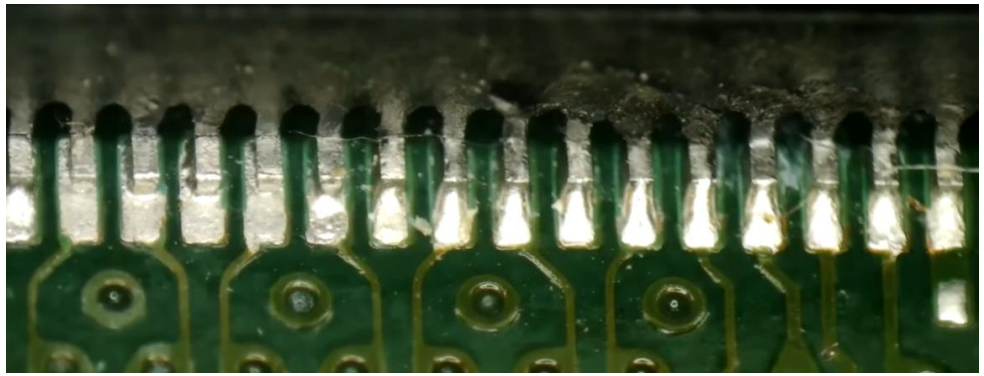
Liquid metal



Check soldering if there is rework done by others

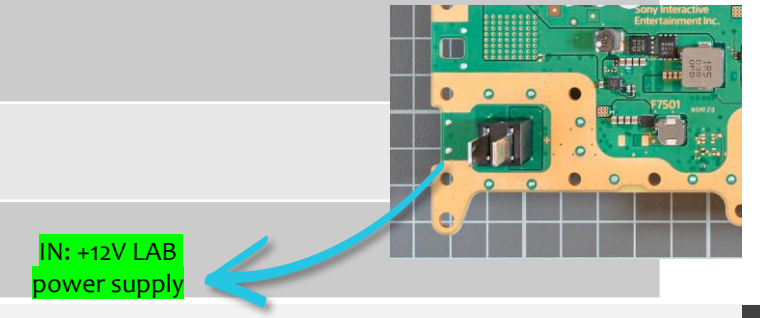
Check if the ic pins / pads are making a good connection with the board. Check if the HDMI port / pins are making a good connection



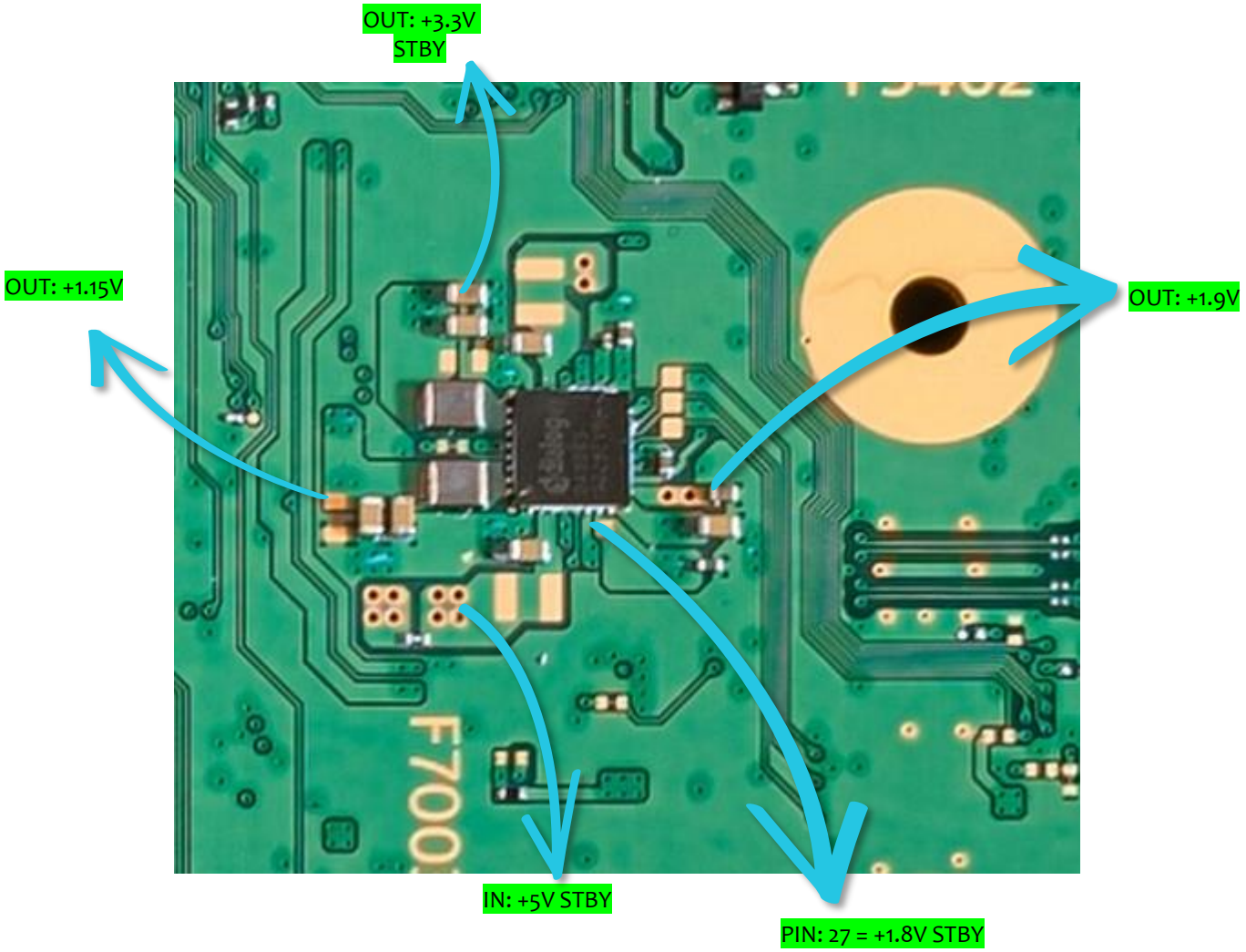


Standby boot sequence – no power button is pressed

First stage	Second stage	Cause
+- 321 mA	+- 7 – 10 mA	WORKING / GOOD
+- 0 mA	+- 0 mA	Check BIOS and fuses, +12V power supply, Check for short on the +12V input
+- 0.015 mA	+- 0.015 mA	Check for shorts near Soutbridge IC
+- 5 mA	+- 5 mA	Check for shorts near HDMI encoder IC
+- 7 mA	+- 27 mA	Check for shorts near Soutbridge IC
+- 7 mA	+- 7 mA	When console get's standby power if it is stuck at 7mA. Check the voltages of Dialog ic and frequency of the 25Mhz clock and the 32Khz clock of the Southbridge IC. If that is working, try to see if the Southbridge is accessing the BIOS IC. Flash firmware / Replace the BIOS IC, Southbridge IC
+- 7 mA (boot loop)	+- 12 mA	Check for short on the 1.1V power rail
+- 24 mA	+- 7 mA	Check fuses and check for shorts
+- 35 mA	+- 35 mA	Check for shorts on the 1.9V power rail
+- 58 mA	+- 7 mA	Check fuses and check for shorts
+- 175 mA	+- 10 mA	Check SSD Controller for shorts
+- 225 mA	+- 7 mA	Check SSD Controller for shorts
+- 270 mA	Current goes down when you press power and you hear a beep	Check for shorts near the Wifi IC
+- 315 mA	+- 315 mA	Check 3.3V STBY power rail present, check for shorts Wifi IC
+- 321 mA	+- 7 – 10mA (it goes fast to sleep)	Check SSD controller. Sometimes you get an update screen or no power behaviour



Dialog / Power Management IC - DA90D65



DA9065 is a power management integrated circuit (PMIC). This IC has different buck converters to generate different power rails.

The most important power rails will go the Southbridge IC.

No datasheet

Block Diagram

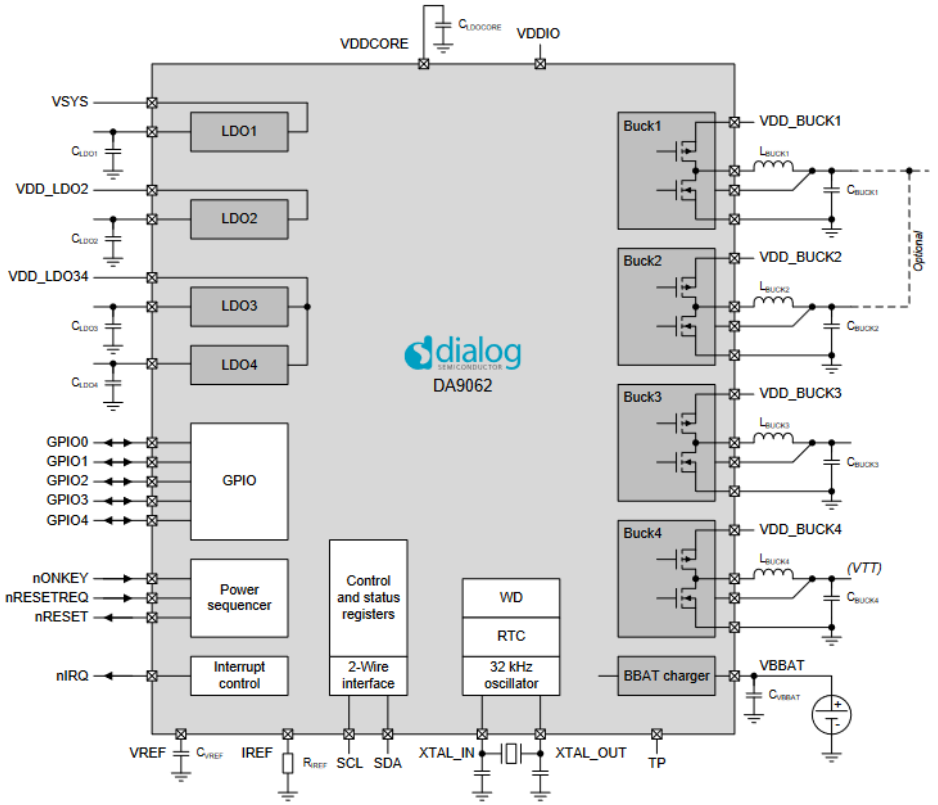
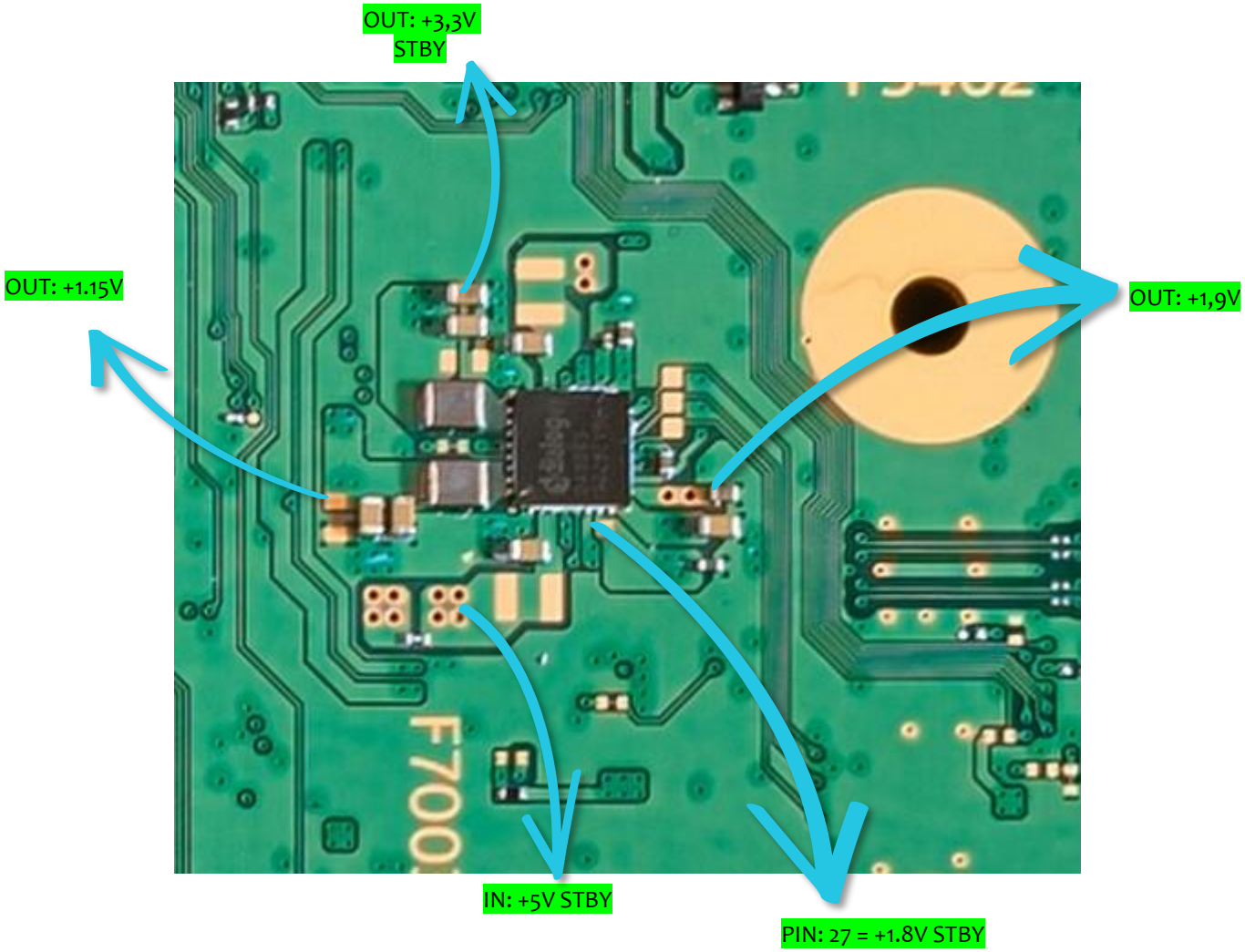


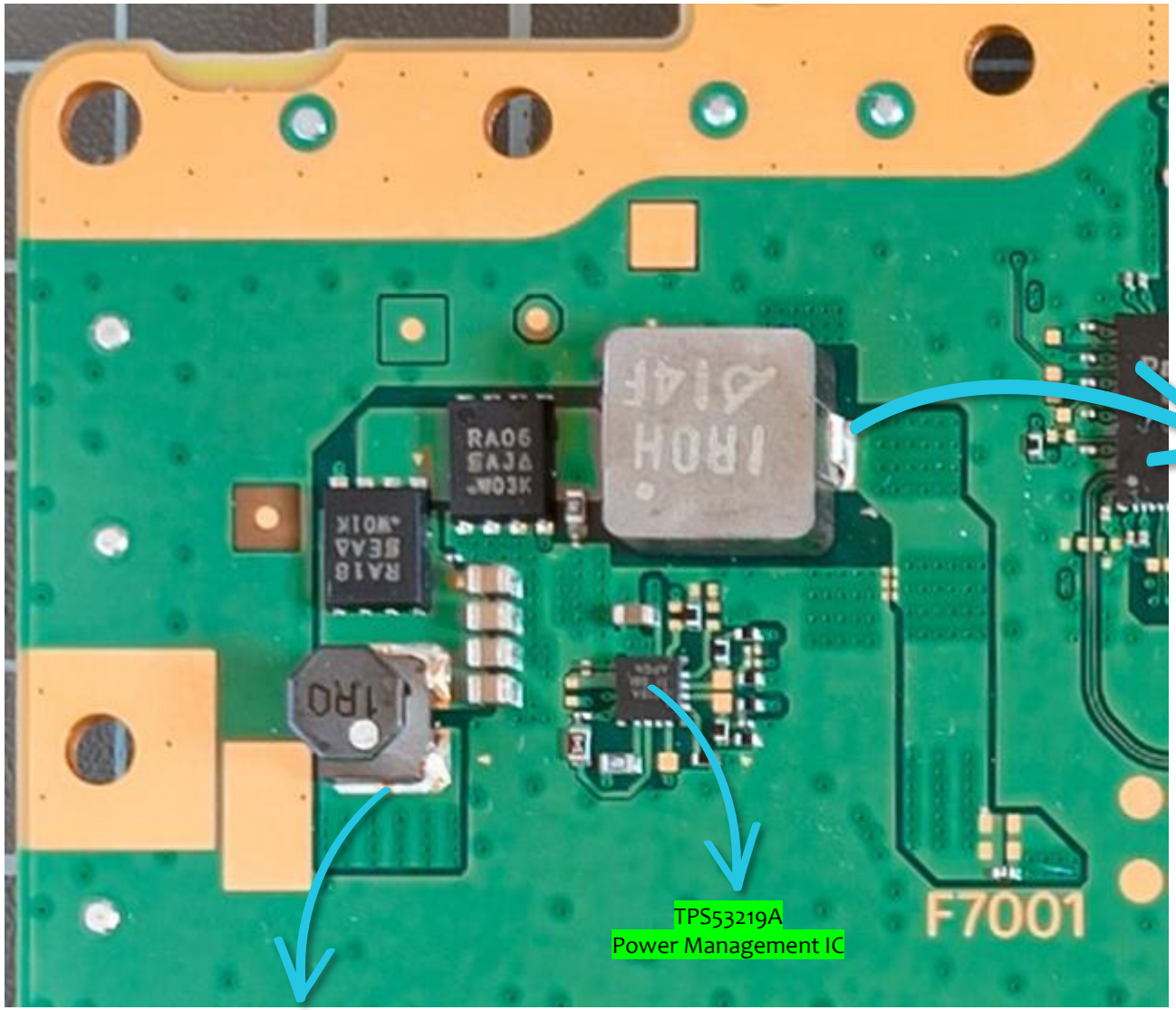
Figure 1: DA9062 Block Diagram

PINOUT - Dialog / Power Management IC - A90D65



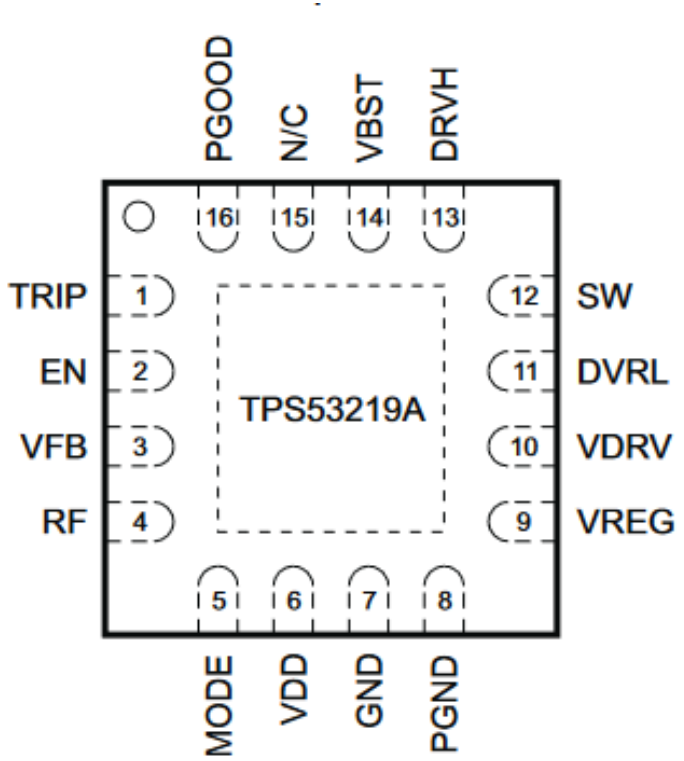
PIN 1 = +- 1.15V
PIN 2 = +- 1.15V
PIN 3 = NC
PIN 5 = +- 5V (F7003)
PIN 6 = NC
PIN 7 = +- 3.3V
PIN 8 = +- 3.3V
PIN 9 = GND
PIN 10 = +- 5V (F7003)
PIN 11 = +- 5V (F7003)
PIN 12 – 16 = +- 3.3V
PIN 17 = +- 5V (F7003)
PIN 21 = GND
PIN 22 = +- 2V
PIN 23 = +- 5V (F7003)
PIN 24 = +- 2V
PIN 25 = +- 1.15V
PIN 26 = +- 3.3V
PIN 28 = +- 3.3V
PIN 27 = +- 1.8V
PIN 29 = +- 1.15V
PIN 30 = Goes to fuse F7003
PIN 31 = Goes to fuse F7003
PIN 32= GND

TPS53219A - single buck controller

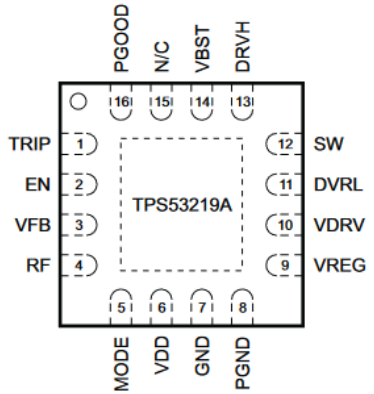


This power regulator will generate +5v STBY. It is created from the +12V power supply.

The +5V STBY is used to create other power rails



TPS53219A - single buck controller



Pin Functions

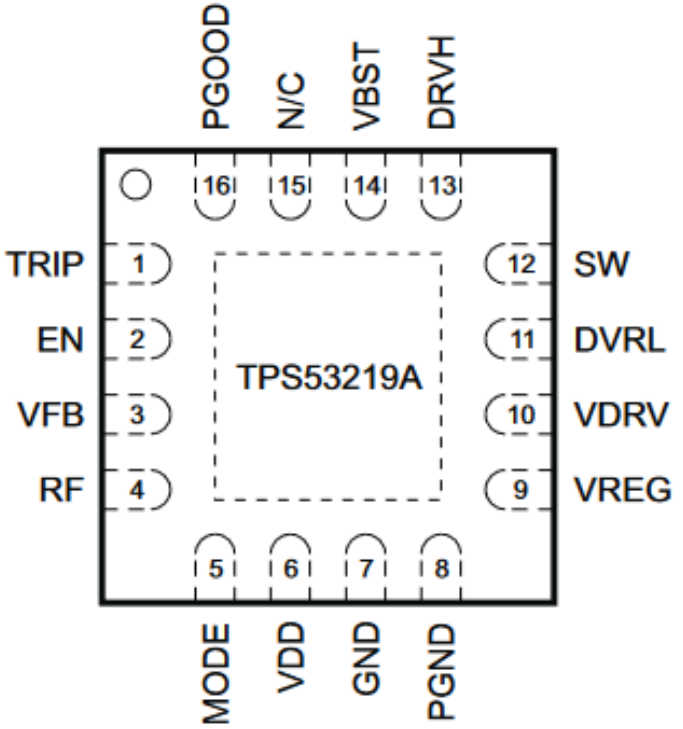
PIN NAME	PIN NO.	TYPE(1)	DESCRIPTION
DRVH	13	O	High-side MOSFET driver output. The SW node referenced floating driver. The gate drive voltage is defined by the voltage across VBST to SW node bootstrap flying capacitor.
DRVL	11	O	Synchronous MOSFET driver output. The PGND referenced driver. The gate drive voltage is defined by VDRV voltage.
EN	2	I	Enable pin. Place a 1-kΩ resistor in series with this pin if the source voltage is higher than 5.5 V.
GND	7	G	Ground pin. This is the ground of internal analog circuitry. Connect to GND plane at single point.
MODE	5	I	Soft-start and skip/CCM selection. Connect a resistor to select soft-start time using Table 1. The soft-start time is detected and stored into internal register during start-up.
NC	15	–	No connection.
PAD	–	–	Thermal pad. Use five vias to connect to GND plane.
PGOOD	16	O	Open-drain power good flag. Provides 1-ms start-up delay after the VFB pin voltage falls within specified limits. When VFB goes out specified limits PGOOD goes low after a 2-µs delay.
PGND	8	G	Power ground. Connect to GND plane.
RF	4	I	Switching frequency selection. Connect a resistor to GND or VREG to select switching frequency using Table 2. The switching frequency is detected and stored during the start-up.
SW	12	P	Output of converted power. Connect this pin to the output inductor.
TRIP	1	I	OCL detection threshold setting pin. 10 µA at room temp, 4700 ppm/°C current is sourced and set the OCL trip voltage as follows. $V_{OCL} = V_{TRIP}/8$ ($V_{TRIP} \le 3$ V, $V_{OCL} \le 375$ mV)
VBST	14	P	Supply input for high-side FET gate driver (boost terminal). Connect a capacitor from this pin to SW-node. Internally connected to VREG through bootstrap MOSFET switch.
VDD	6	P	Controller power supply input. The input range is from 4.5 V to 25 V.
VDRV	10	I	Gate drive supply voltage input. Connect to VREG if using LDO output as gate drive supply.
VFB	3	I	Output feedback input. Connect this pin to V_{OUT} through a resistor divider.
VREG	9	O	6.2-V LDO output. This is the supply of internal analog circuitry and driver circuitry.

(1) I=Input, O=Output, P=Power, G=Ground

Most important pins to measure

This power regulator will generate +5v STBY. It is created from the +12V power supply.

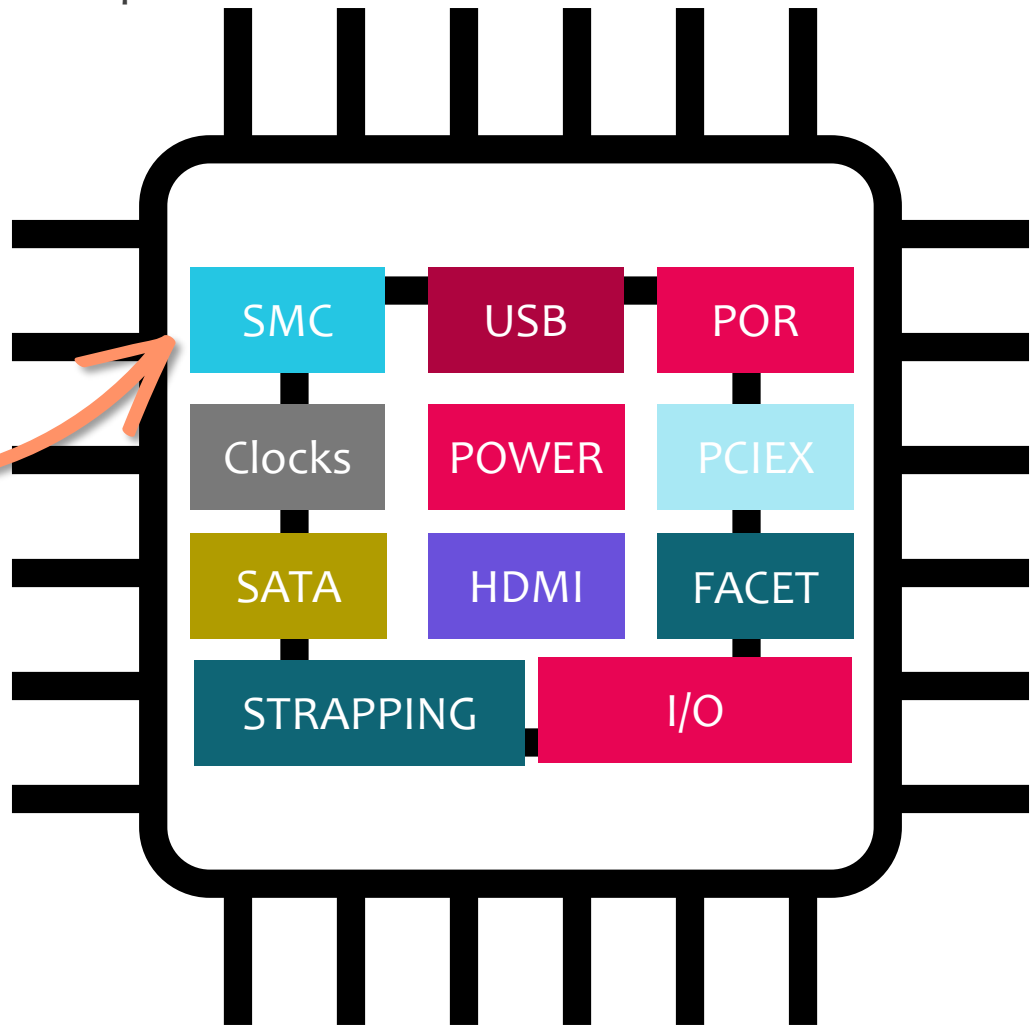
The +5V STBY is used to create other power rails



Southbridge

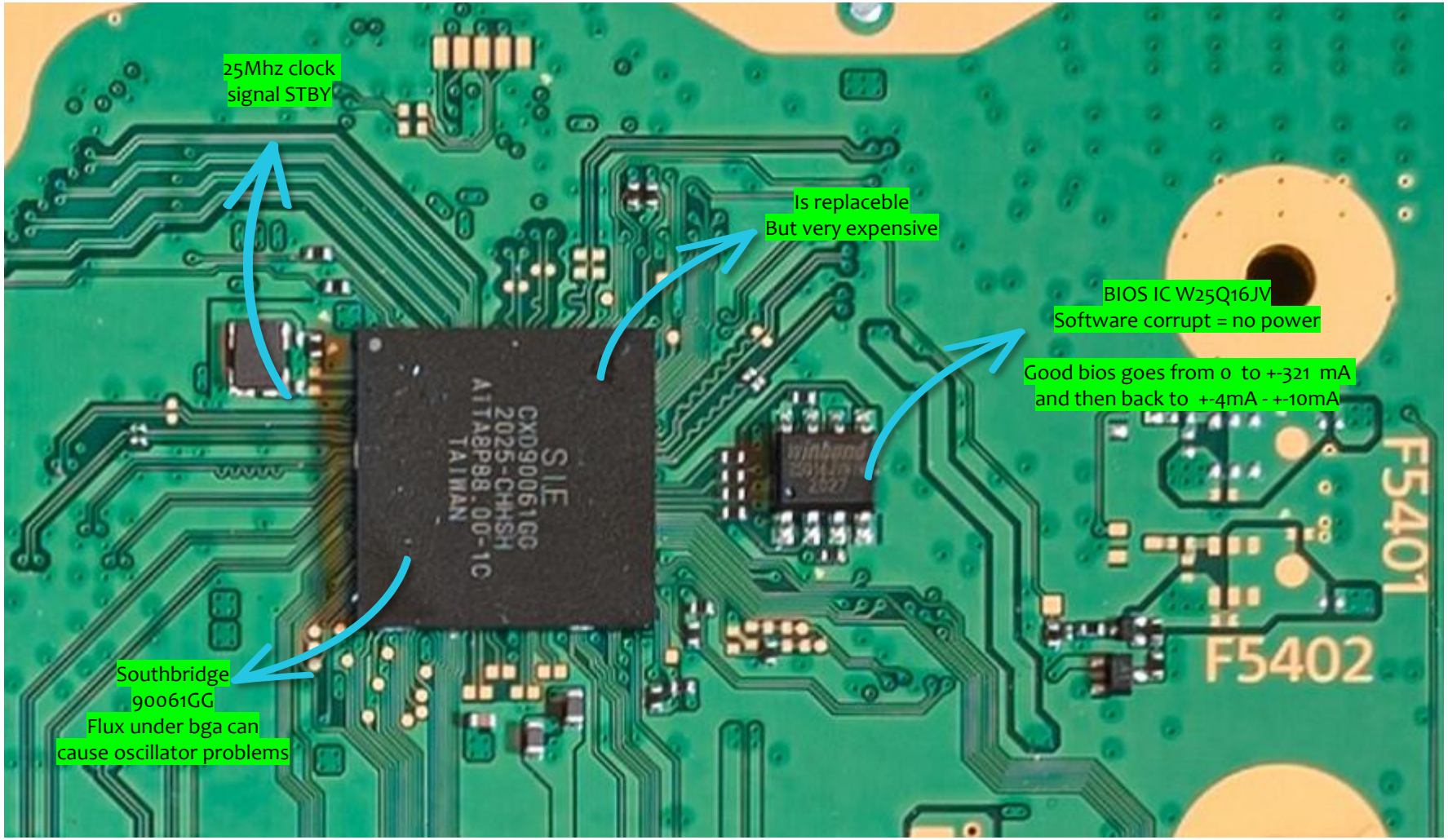
All I/O operations are handled through a chip called Southbridge The Southbridge connects to all external interfaces. This chip is connected to the SOC chip.

The SMC is responsible for all I/O operations, power management, real-time clock, temperature control, LED control





Southbridge IC – CXD90061 and BIOS IC - W25Q16JVSNIM



Southbridge IC is an expensive IC.
Price +- 50$

There are two crystal oscillator
25Mhz and a 32Khz

Flux under BGA can cause oscillator problems.

If 3.3V STBY is missing the power buttons will not work. Southbridge, Encoder IC, a switching regulator, Dialog Power management IC, WiFi IC

If you press power button the power rail +1.1V must be generated for the Southbridge IC.

When console **beep 3 times** but not powers on check all voltages if all ok it could be a BIOS corruption.

BIOS IC - W25Q16JVSNIM

3.1 Pin Configuration SOIC 150/208-mil

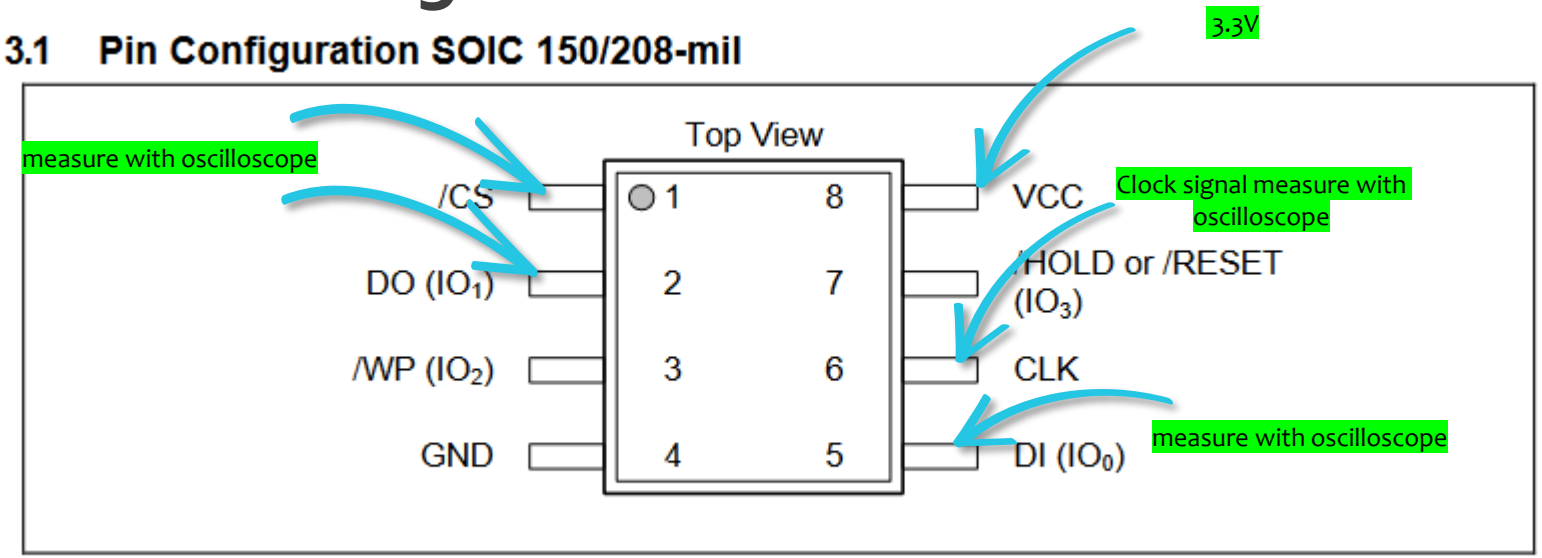


Figure 1a. W25Q16JV Pin Assignments, 8-pin SOIC 150/208-mil (Package Code SN, SS)

3.3 Pin Description SOIC 150/208-mil, WSON 6x5-mm, USON 2x3-mm/4x3-mm

PIN NO.	PIN NAME	I/O	FUNCTION
1	/CS	I	Chip Select Input
2	DO (IO1)	I/O	Data Output (Data Input Output 1)(1)
3	/WP (IO2)	I/O	Write Protect Input (Data Input Output 2)(2)
4	GND		Ground
5	DI (IO0)	I/O	Data Input (Data Input Output 0)(1)
6	CLK	I	Serial Clock Input
7	/HOLD or /RESET (IO3)	I/O	Hold or Reset Input (Data Input Output 3)(2)
8	VCC		Power Supply

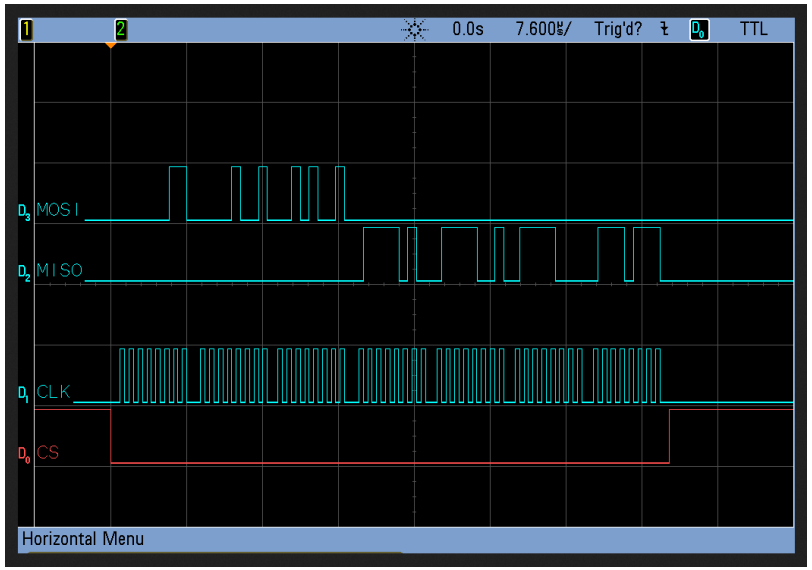
2.7V to 3.6V

Check bios pins with oscilloscope and see if there is communication going into the BIOS and also out of the BIOS.

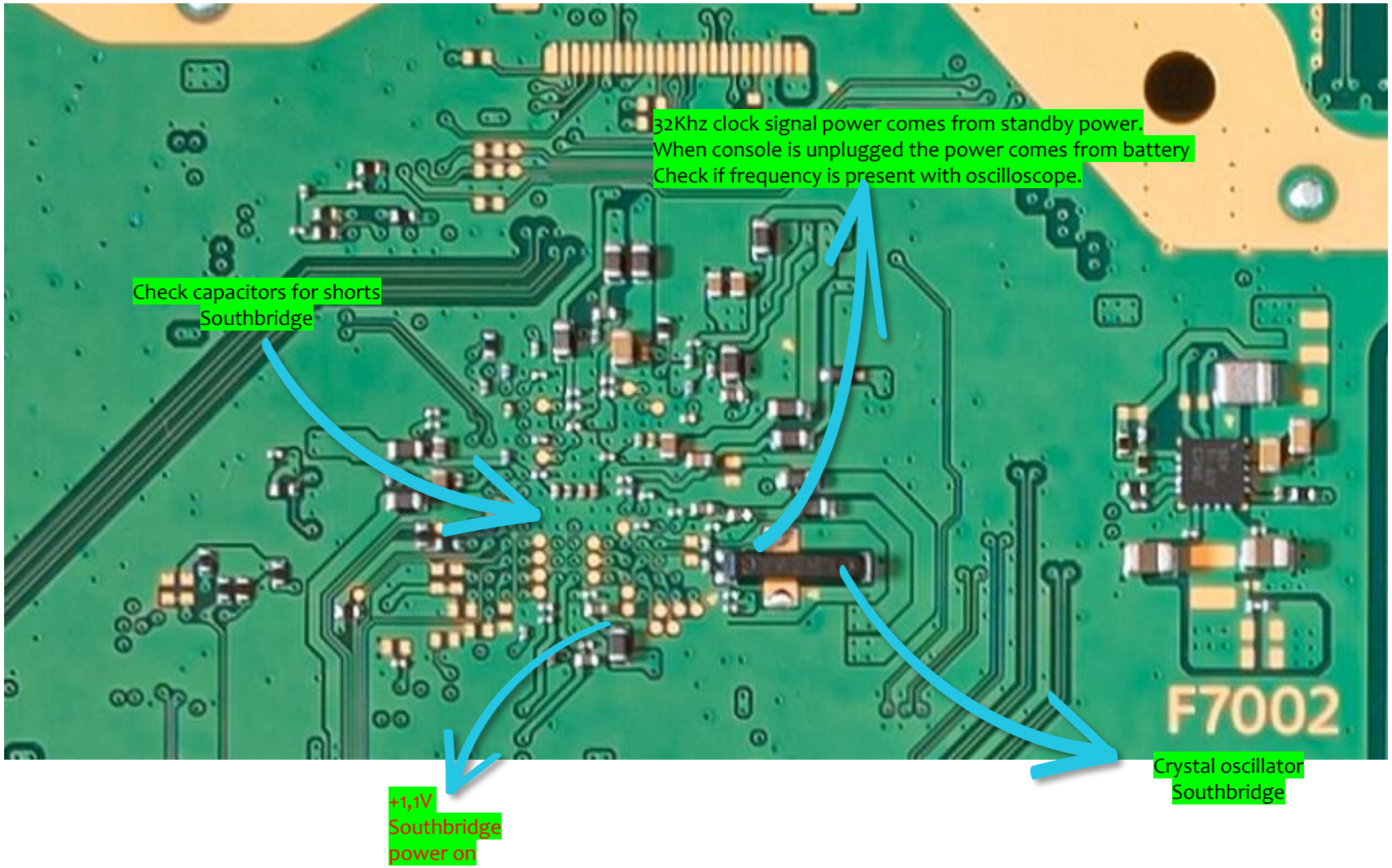
Check if the clock signal is present.

If communication is present at the DI pin but there is no communication going out at pin DO then firmware / IC could be faulty.

Test signals in standby mode and power on mode.



Soutbridge IC – back side



Southbridge IC is an expensive IC.
Price +- 50$

There are two crystal oscillator
25Mhz and a 32Khz

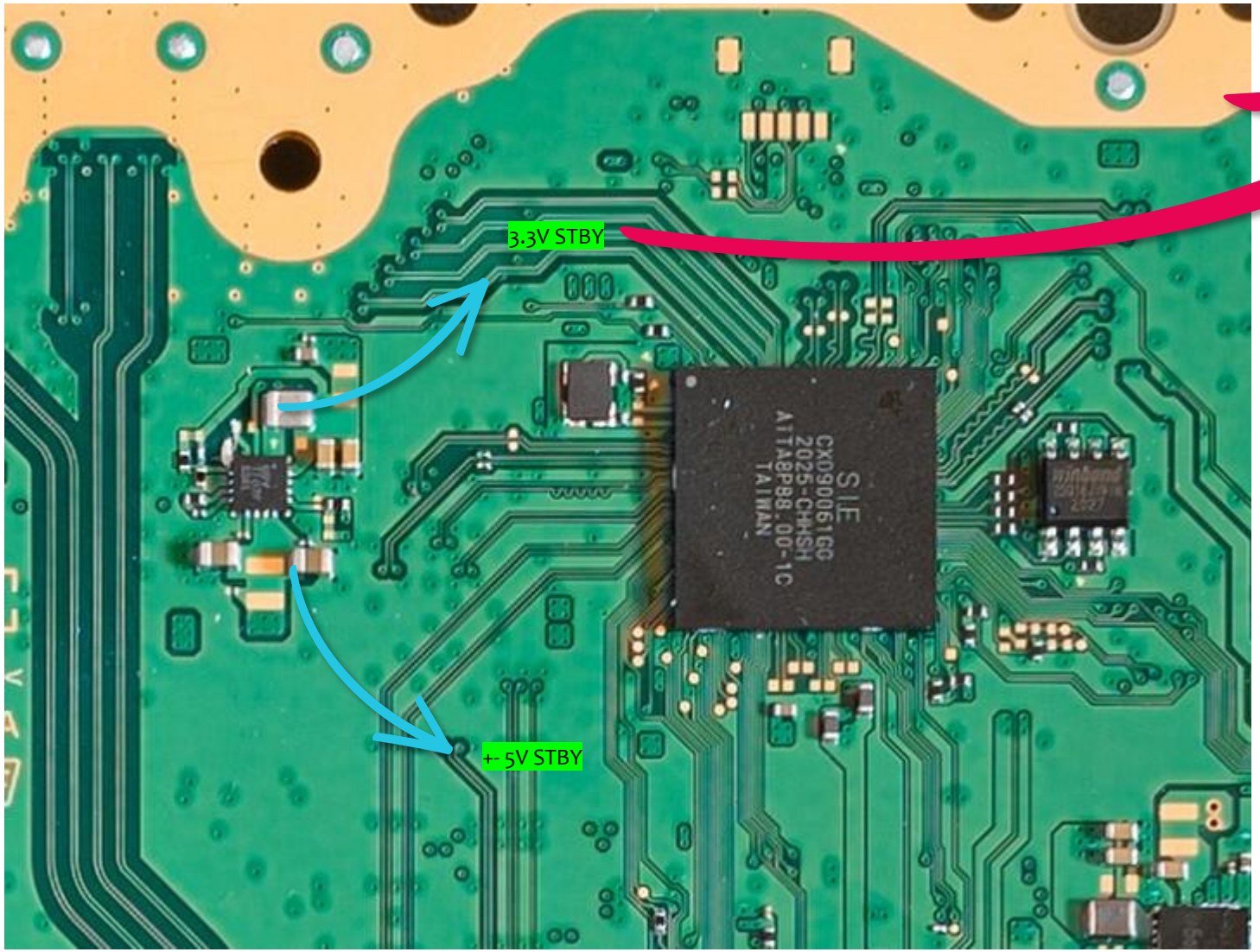
Flux under BGA can cause oscillator problems.

If 3.3V STBY is missing the power buttons will not work. Southbridge, Encoder IC, a switching regulator, Dialog Power management IC, WiFi IC

If you press power button the power rail +1.1V must be generated for the Southbridge IC.

When console **beep 3 times** but not powers on check all voltages if all ok it could be a BIOS corruption.

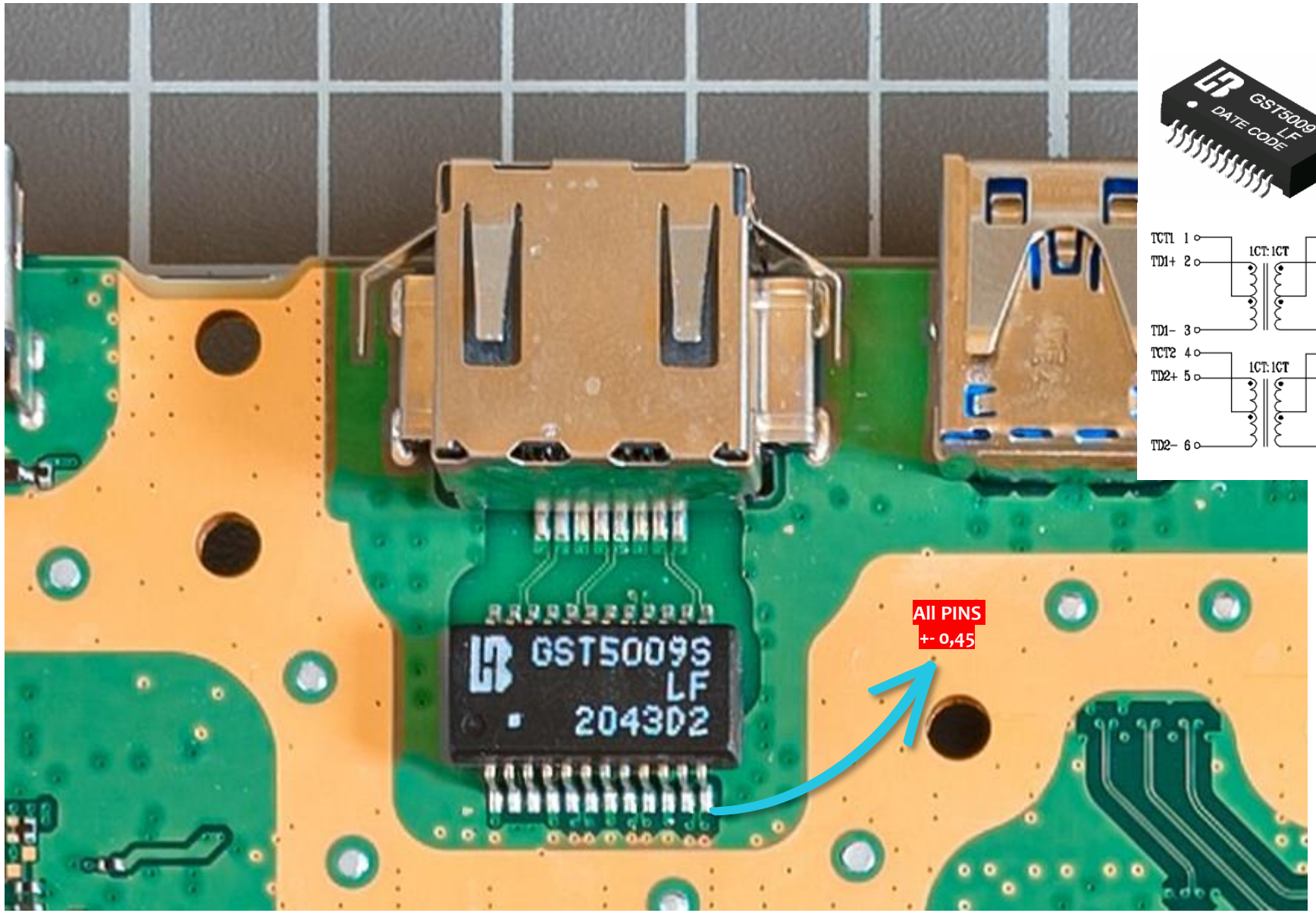
Voltage regulator

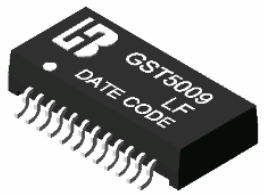


If 3.3V STBY is missing the power buttons will not work.

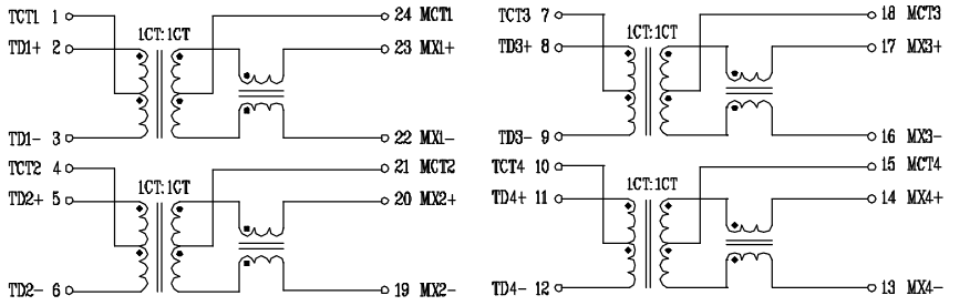
Southbridge, Encoder IC, a switching regulator, Dialog Power management IC, WiFi IC needs 3.3V to operate.

Ethernet port



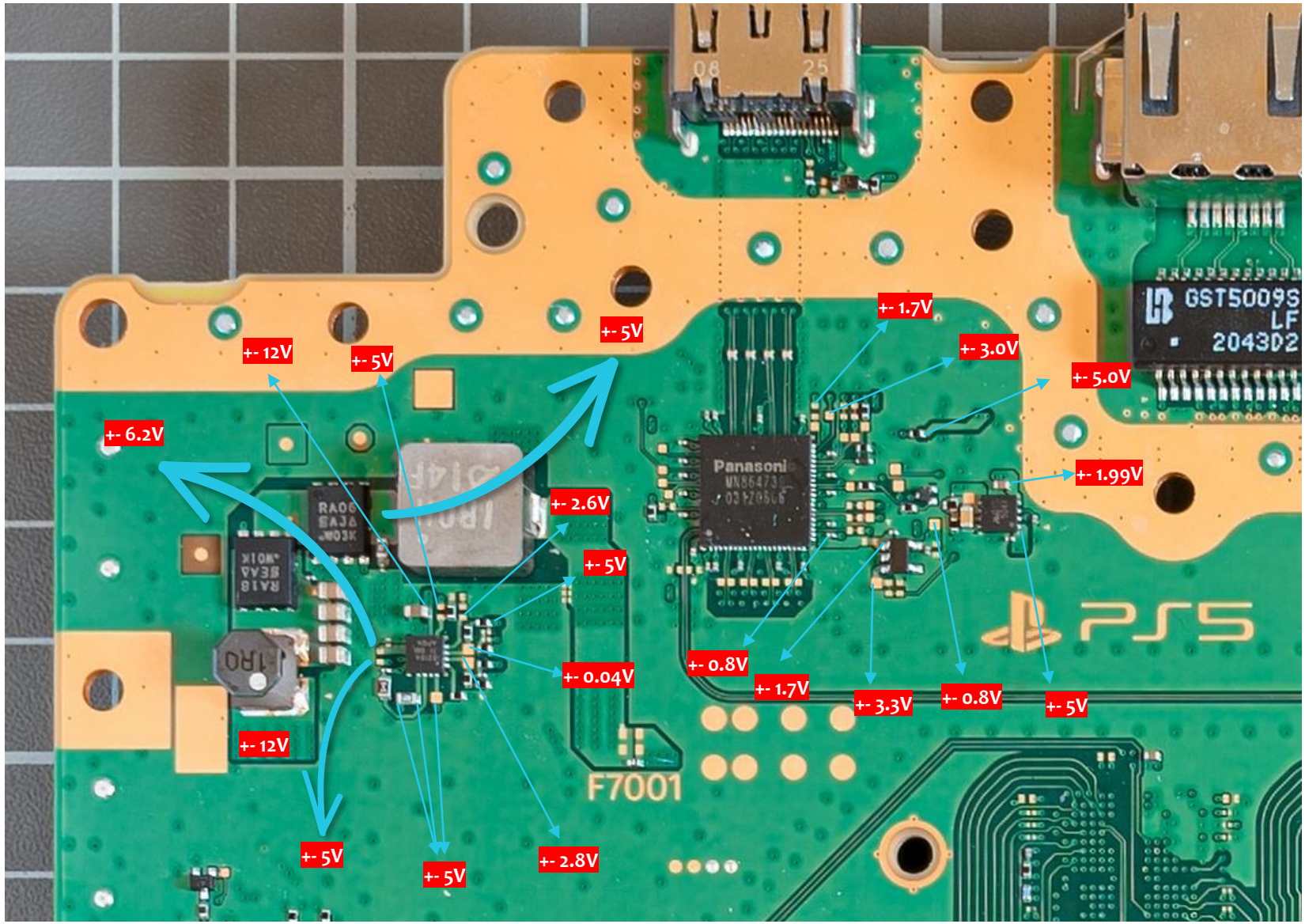


- Designed for long haul gigabit Ethernet 1000 BASE-T, full duplex applications.
- Supports 4 pairs of category 5 UTP cable.
- Cable interface for isolation and low common mode emissions.
- Compliant with IEEE 802.3ab standard for 1000 BASE-T.
- Comply with RoHS requirements-whole part No Cd, No Hg, No Cr6+, No PBB and PBDE and No Pb on external pins.
- Operating temperature range: 0℃ to +70℃.
- Storage temperature range: -25℃ to +125℃.

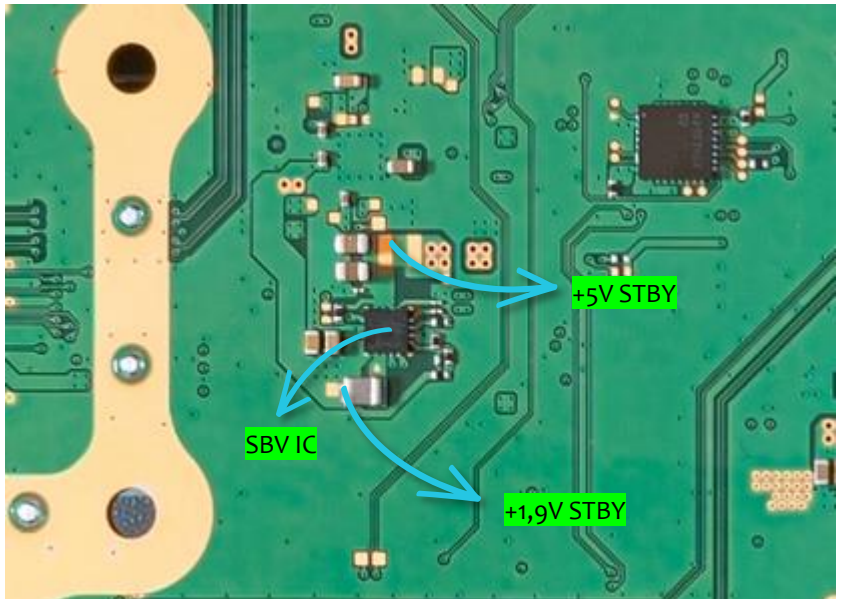


DIODE MEASUREMENT

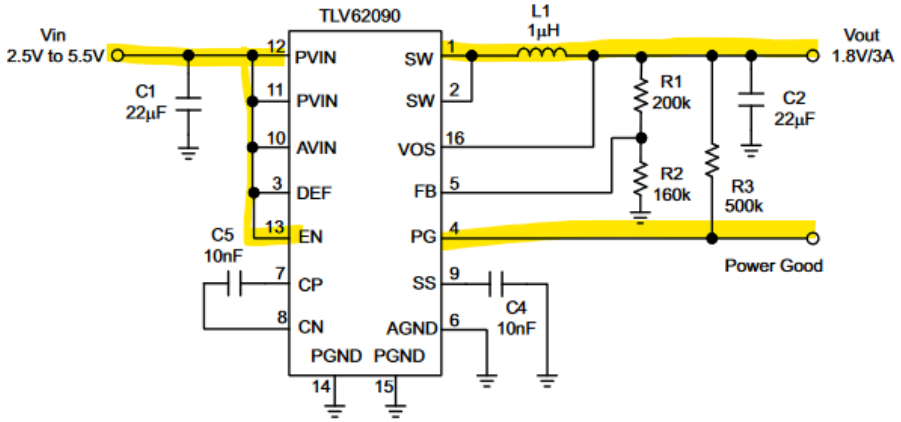
Standby voltages near HDMI encoder



Voltage regulator - TLV62090RGTT - SBV IC - STBY

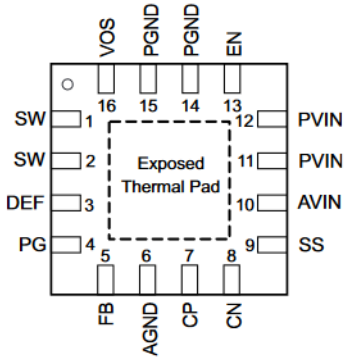


Typical Application Schematic



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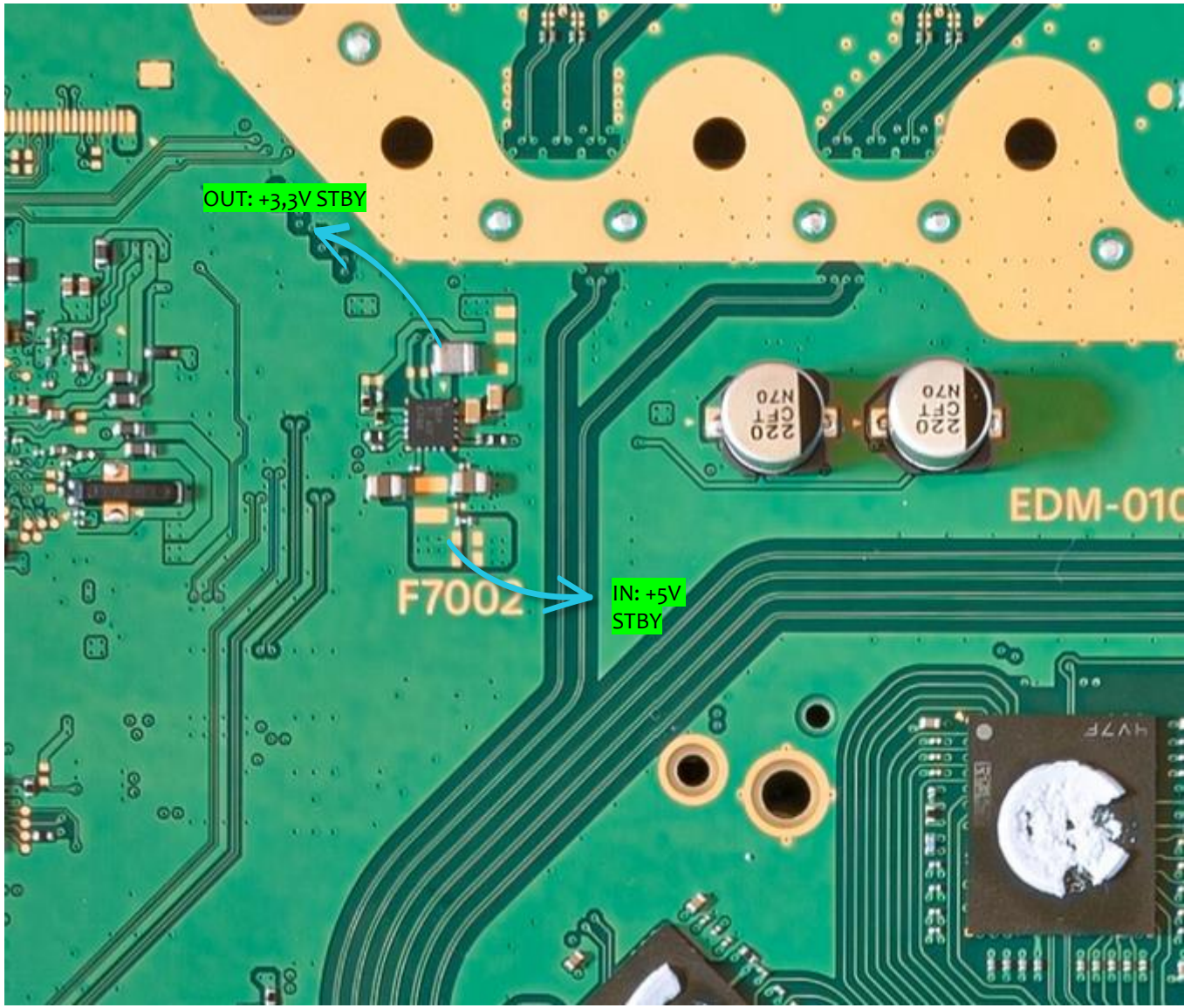
RGT Package
16-Pin VQFN
Top View



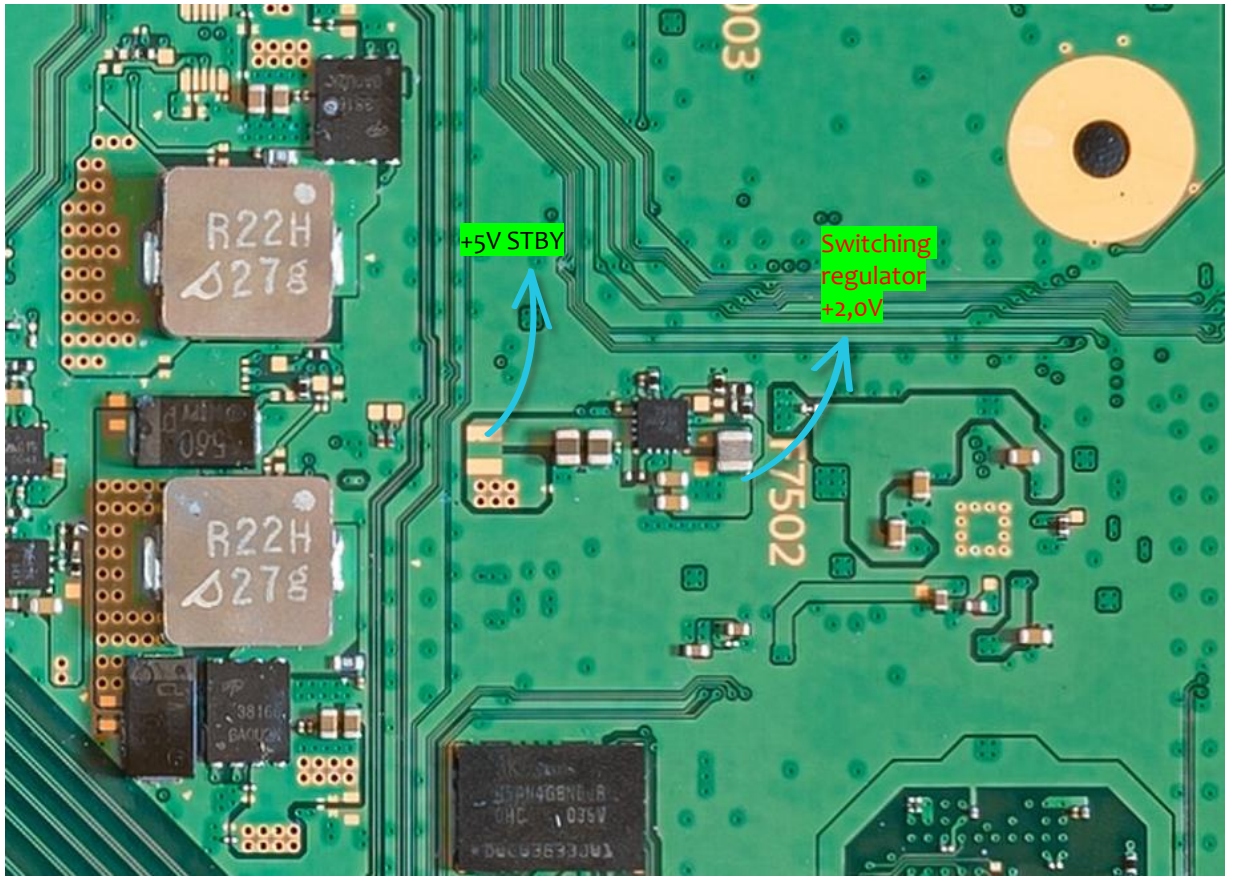
Pin Functions

PIN		I/O	DESCRIPTION
NAME	NO.		
SW	1, 2	I/O	Switch pin of the power stage.
DEF	3	I	This pin is used for internal logic and needs to be pulled high. This pin should not be left floating.
PG	4	O	Power good open drain output. This pin is high impedance if the output voltage is within regulation. This pin is pulled low if the output is below its nominal value. The pull up resistor can not be connected to any voltage higher than the input voltage of the device.
FB	5	I	Feedback pin of the device. Connect a resistor divider to set the output voltage.
AGND	6		Analog ground.
CP	7	I/O	Internal charge pump flying capacitor. Connect a 10 nF capacitor between CP and CN.
CN	8	I/O	Internal charge pump flying capacitor. Connect a 10 nF capacitor between CP and CN.
SS	9	I	Softstart control pin. A capacitor is connected to this pin and sets the softstart time. Leaving this pin floating sets the minimum start-up time.
AVIN	10	I	Bias supply input voltage pin.
PVIN	11,12	I	Power supply input voltage pin.
EN	13	I	Device enable. To enable the device this pin needs to be pulled high. Pulling this pin low disables the device. This pin has a pull down resistor of typically 400 kΩ, which is active when EN is low.
PGND	14,15		Power ground connection.
VOS	16	I	Output voltage sense pin. This pin needs to be connected to the output voltage.
Exposed Thermal Pad			The exposed thermal pad is connected to AGND. It must be soldered for mechanical reliability.

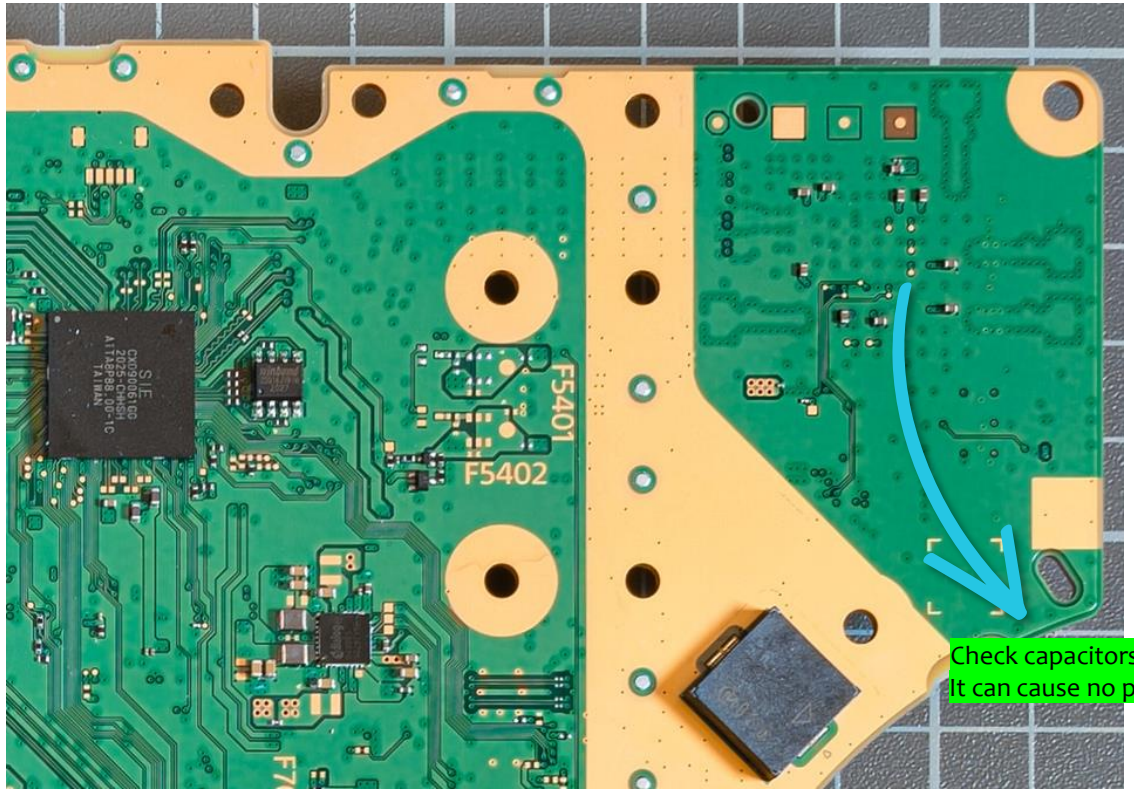
Voltage regulator +5 V to 3.3V



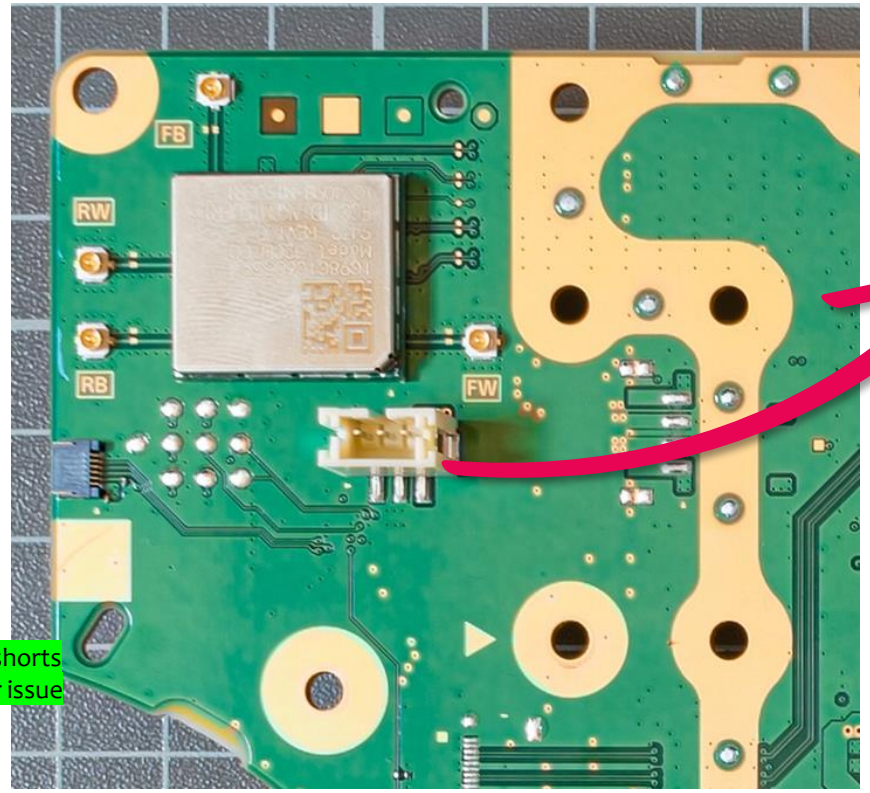
Standby and power voltages

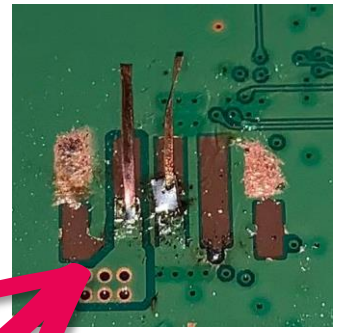


WiFi check for shorts – Standby power



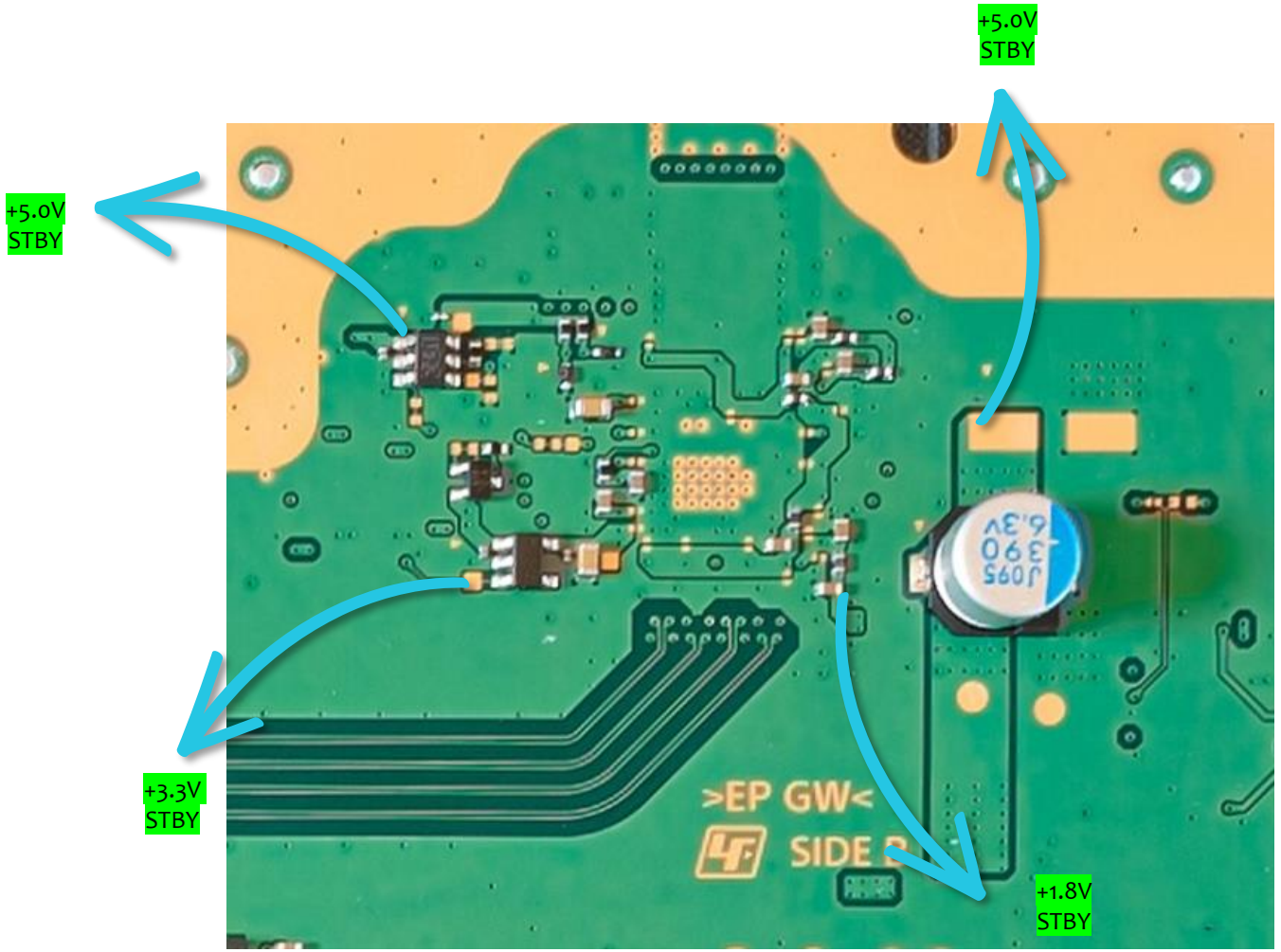
Check capacitors for shorts
It can cause no power issue



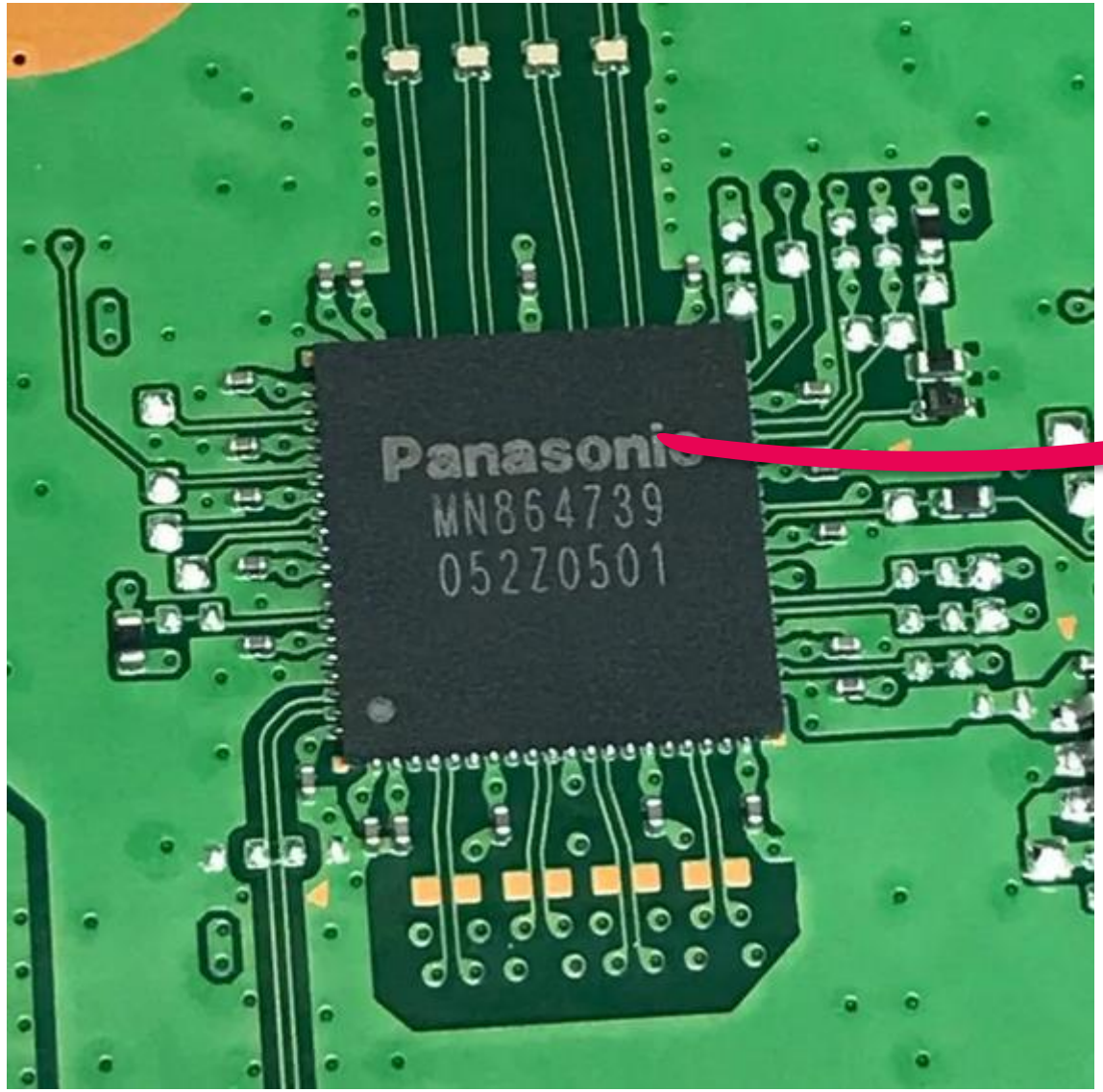


Check for ripped pads

Standby power



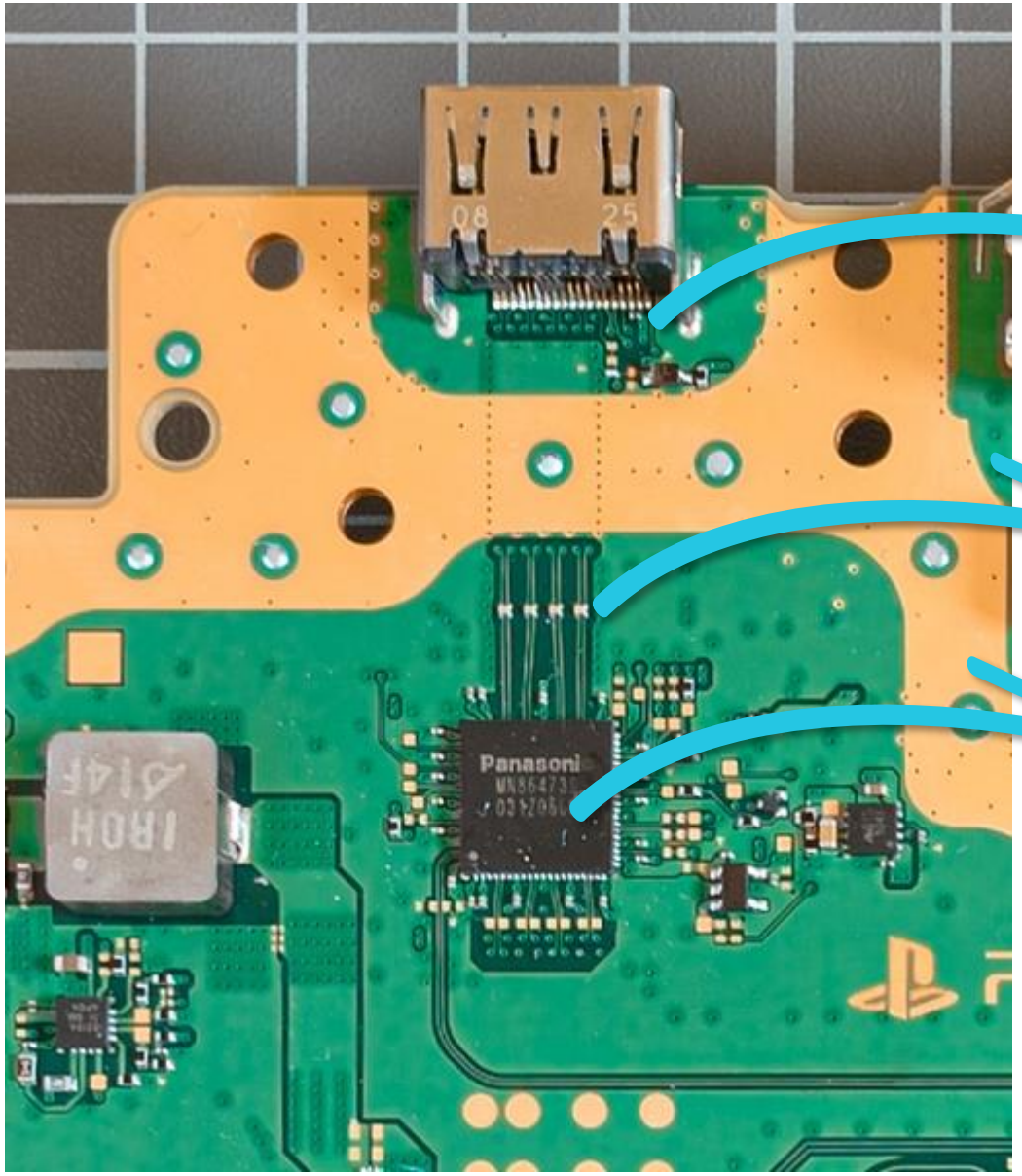
HDMI Encoder IC



If the Encoder IC fails, you can get

- No Picture
- No Power
- blue light of death
- Also 2 seconds of death

White light of death – HDMI - Power on



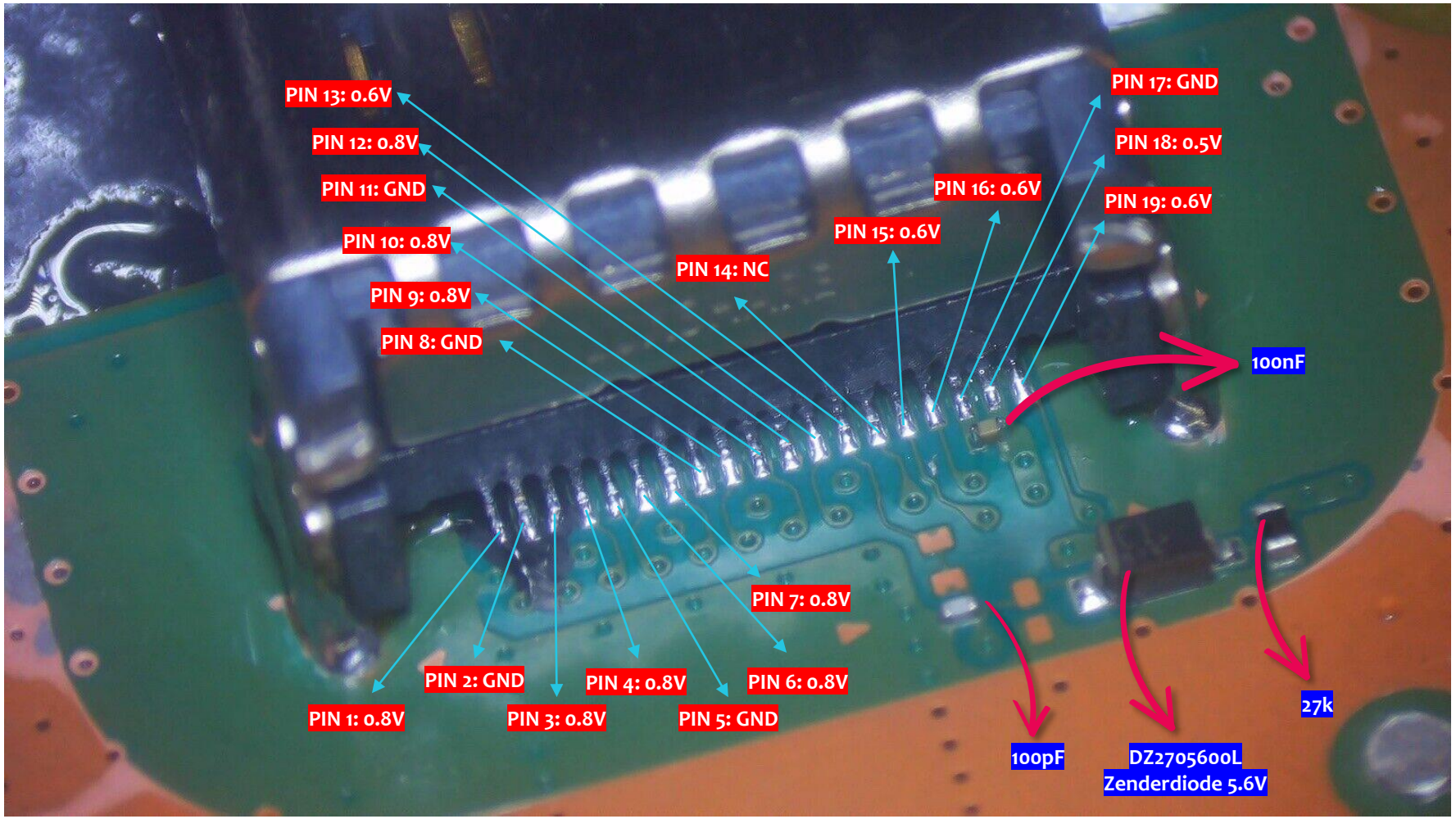
Check HDMI port
and
Nearby components

Check data lines and filters

Check encoder IC

fixmyapp
Research

HDMI PIN DIODE MEASUREMENTS



DIODE MEASUREMENT

Blue Light Of Dead



Long pulsing blue light of dead are hard to solve.

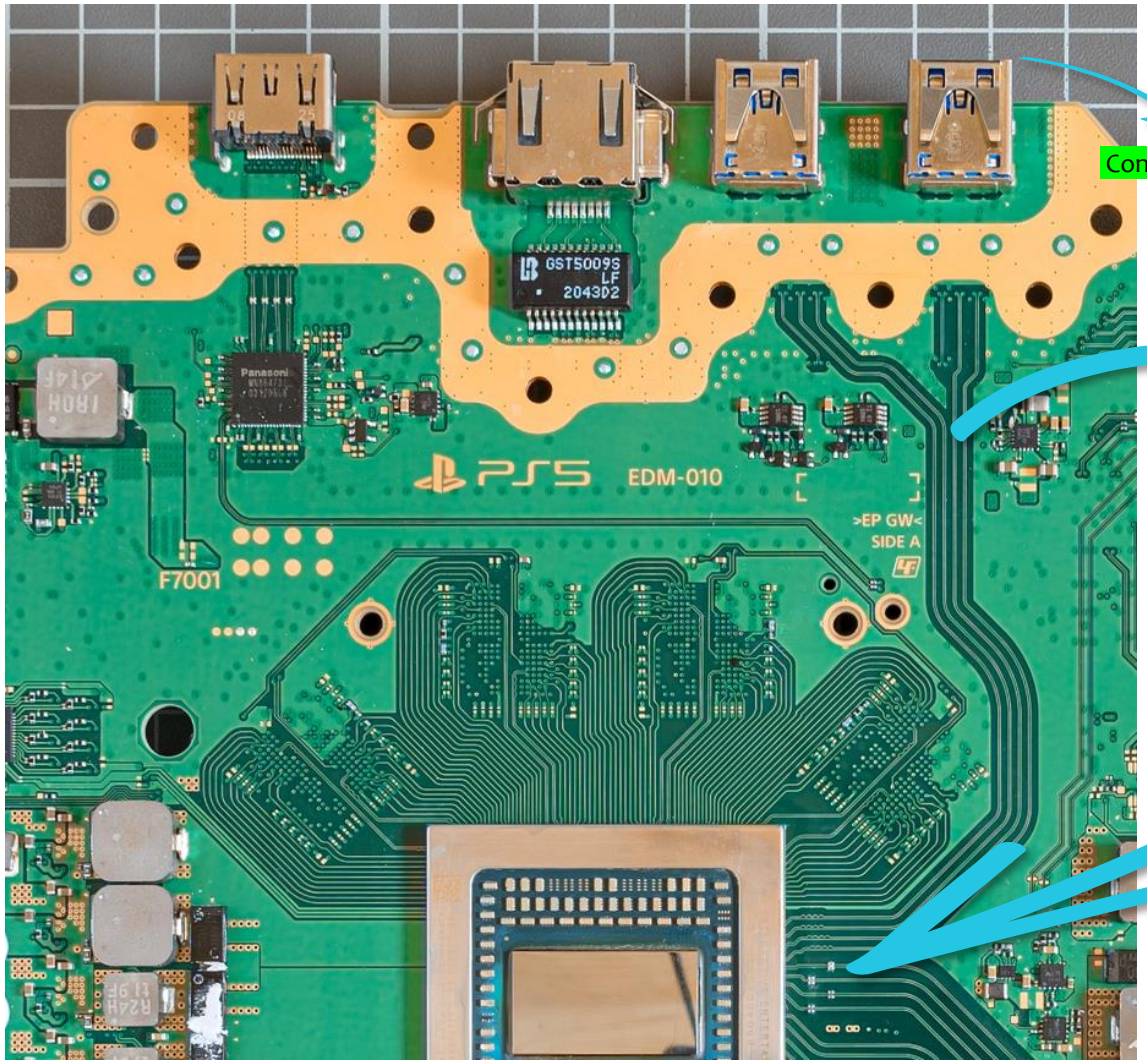
Check board with thermal camera to check for shorts. Problem could be anything. It is like a generic error code. Is keeps pulsating because it keeps running checks.

Blue light and after 2 second it turns off is a common error of PS5. Could be anything.

- First check liquid metal spills on board and APU. Check APU if metal liquid is under the seals or under the APU.
- Probe board in diode mode to find shorts. If all power rails are present and no shorts found, then it can be a BGA IC problem. Most likely it is APU / RAM problem.
- Check with thermal camera to find abnormalities on board.

1 beep every 5-10 seconds when pressing power button can indicate a short.

Update firmware with a USB stick gives a error message



Connect USB drive to update console

If the error message is:

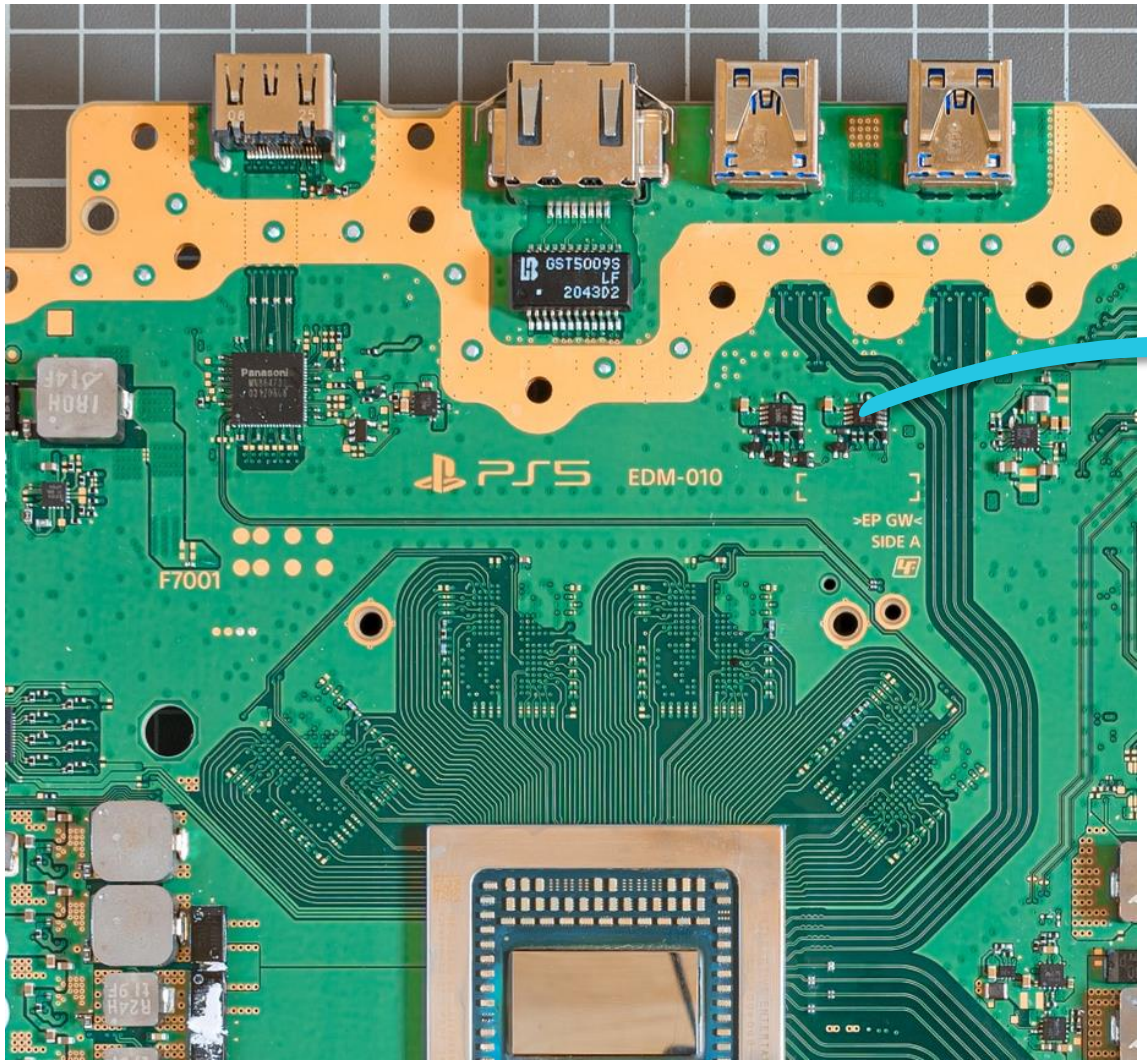
Can't perform this action.

Multiple USB drives are connected

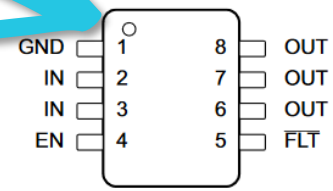
APU is faulty and not fixable

USB data lines go direct to the APU

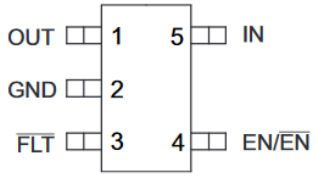
USB - TPS2001D Current Limited, Power-Distribution Switches



DGK Package
8-Pin VSSOP
Top View



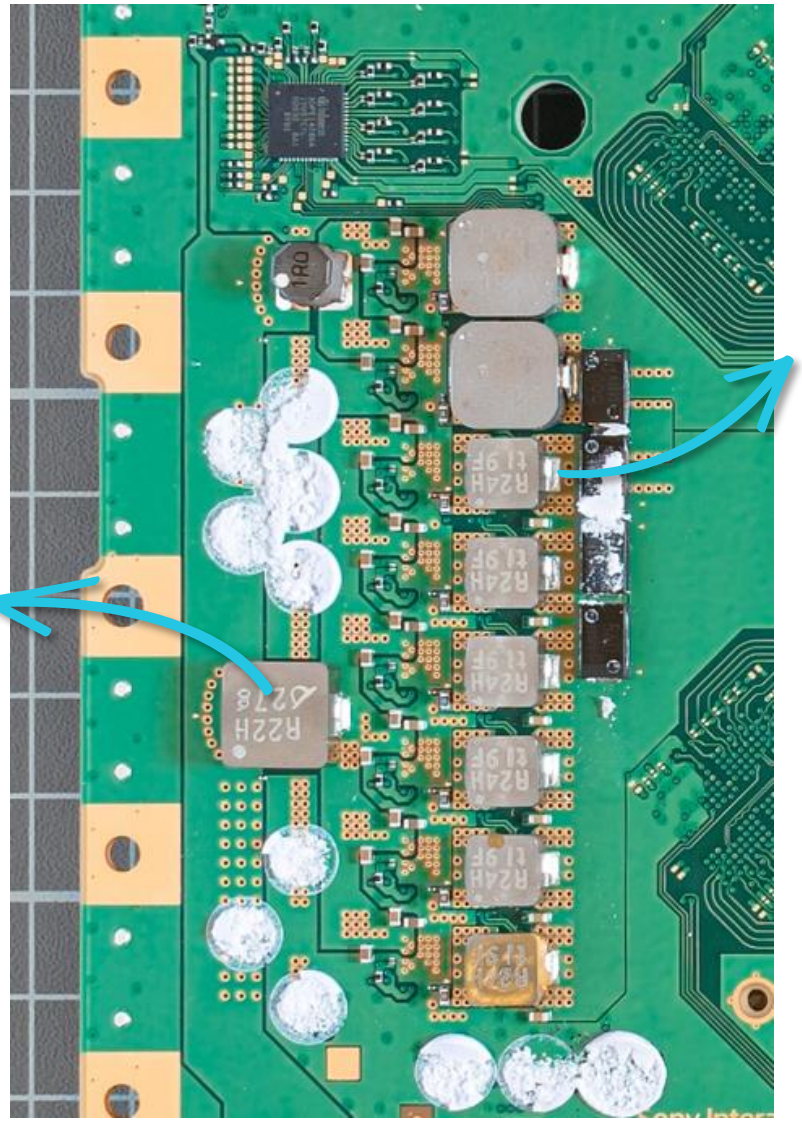
DBV Package
5-Pin SOT-23
Top View

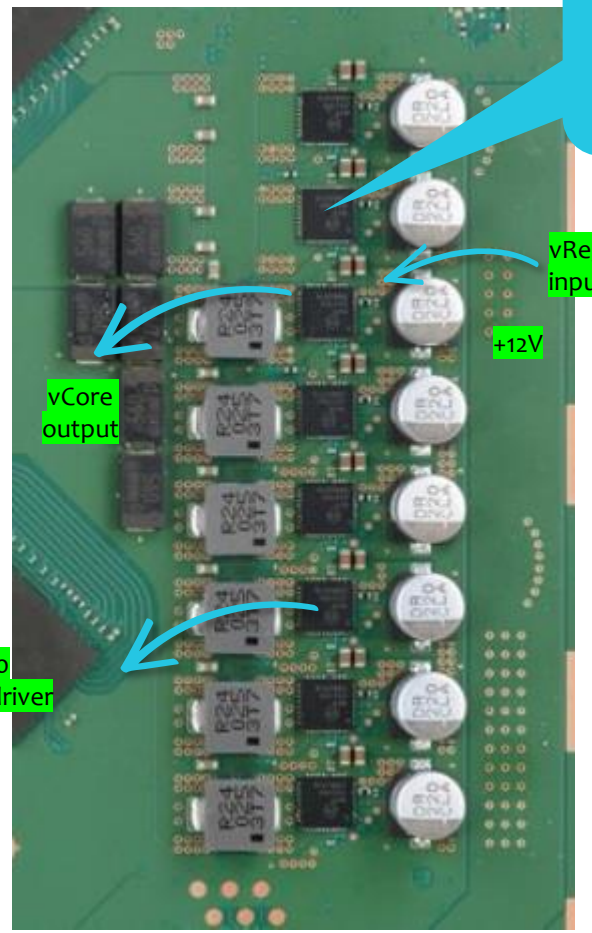


Pin Functions - DGK Package

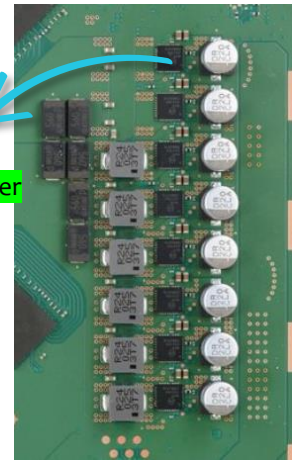
PIN		I/O	DESCRIPTION
NAME	NO.		
EN	4	I	Enable input, logic high turns on power switch
FLT	5	O	Active-low open-drain output, asserted during overcurrent, or overtemperature conditions
GND	1	—	Ground connection
IN	2, 3	PWR	Input voltage and power-switch drain; connect a 0.1-µF or greater ceramic capacitor from IN to GND close to the IC
OUT	6, 7, 8	PWR	Power-switch output, connect to load

+12V power supply to +1.0 Vcore - Power on





NCP252160 - VRM mosfet driver



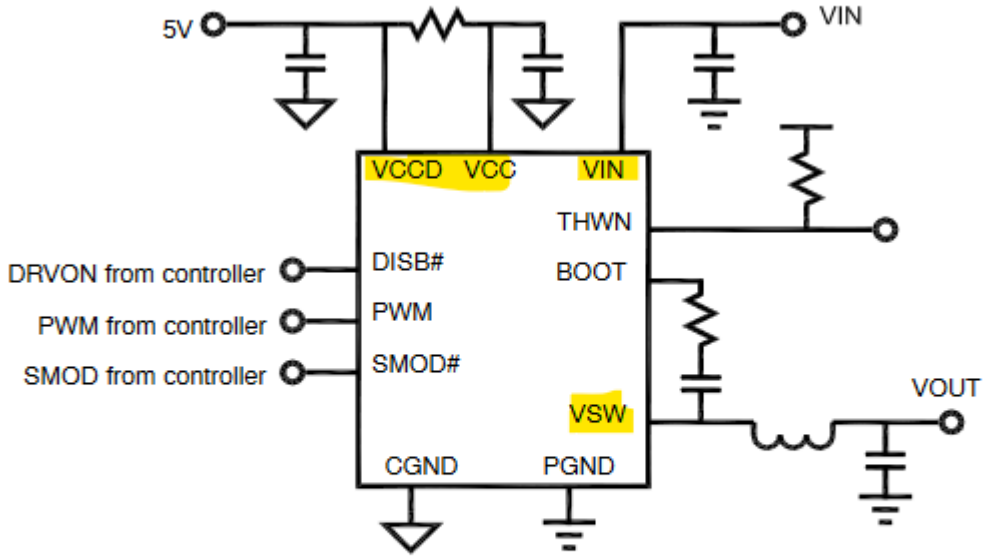
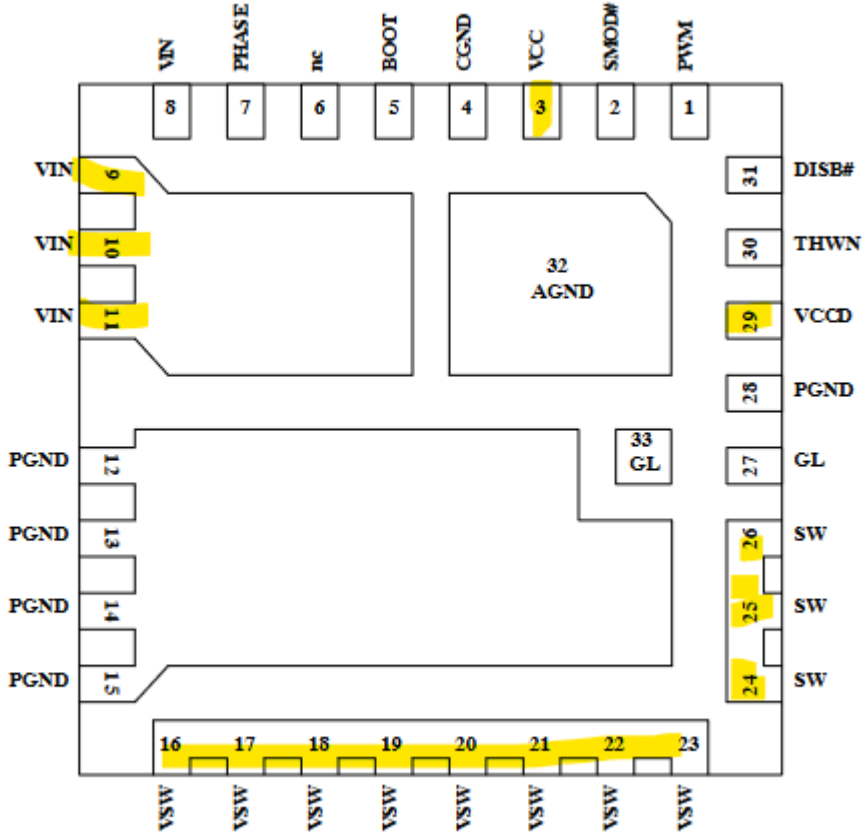
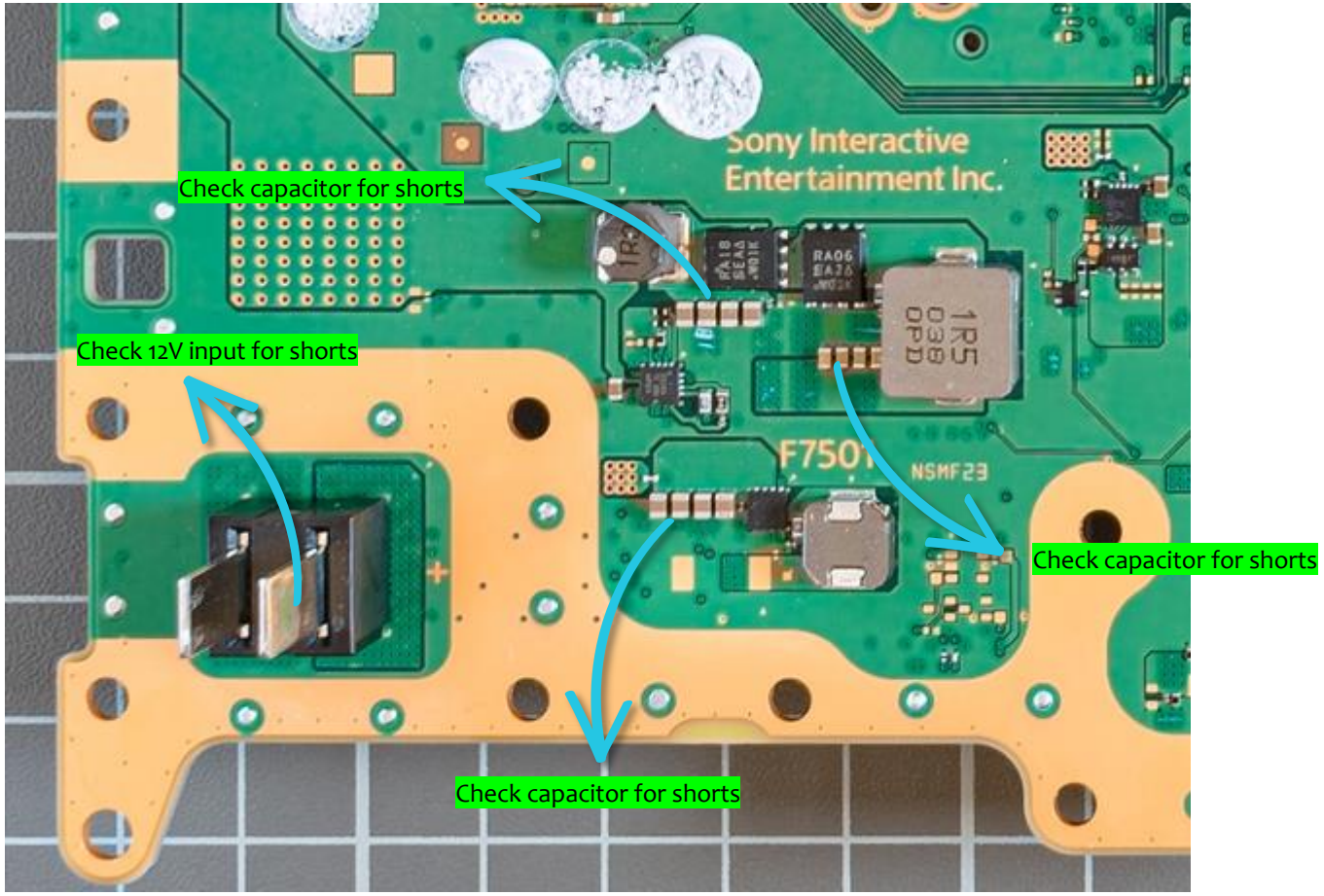


Figure 1. Application Schematic

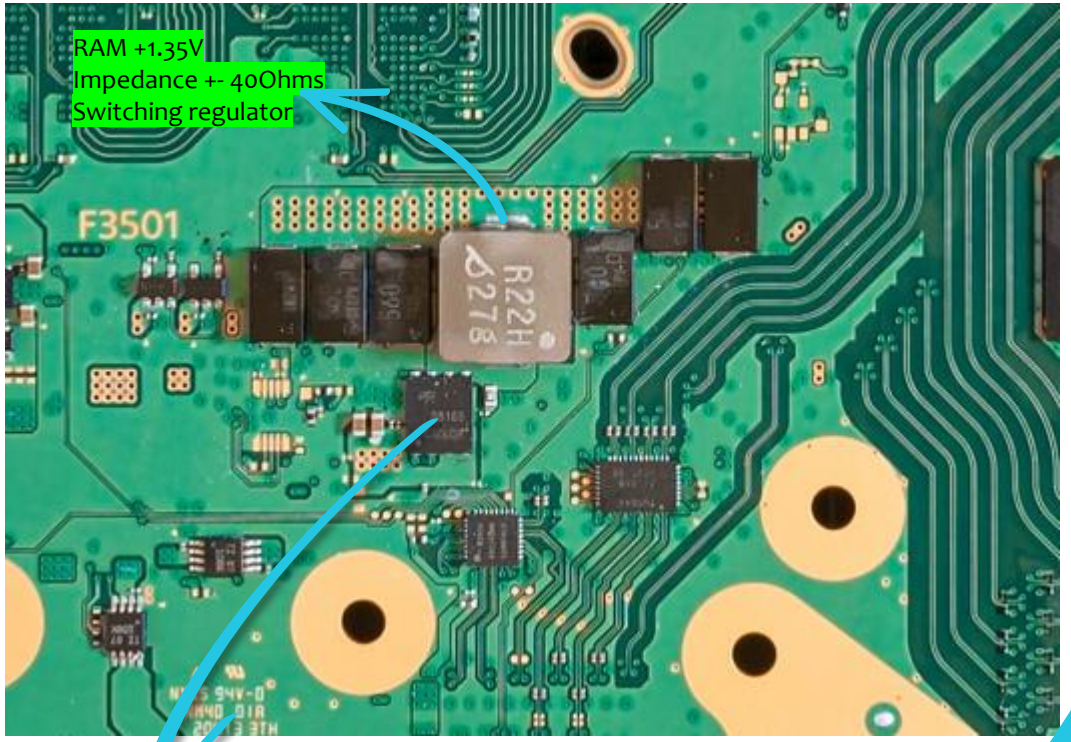


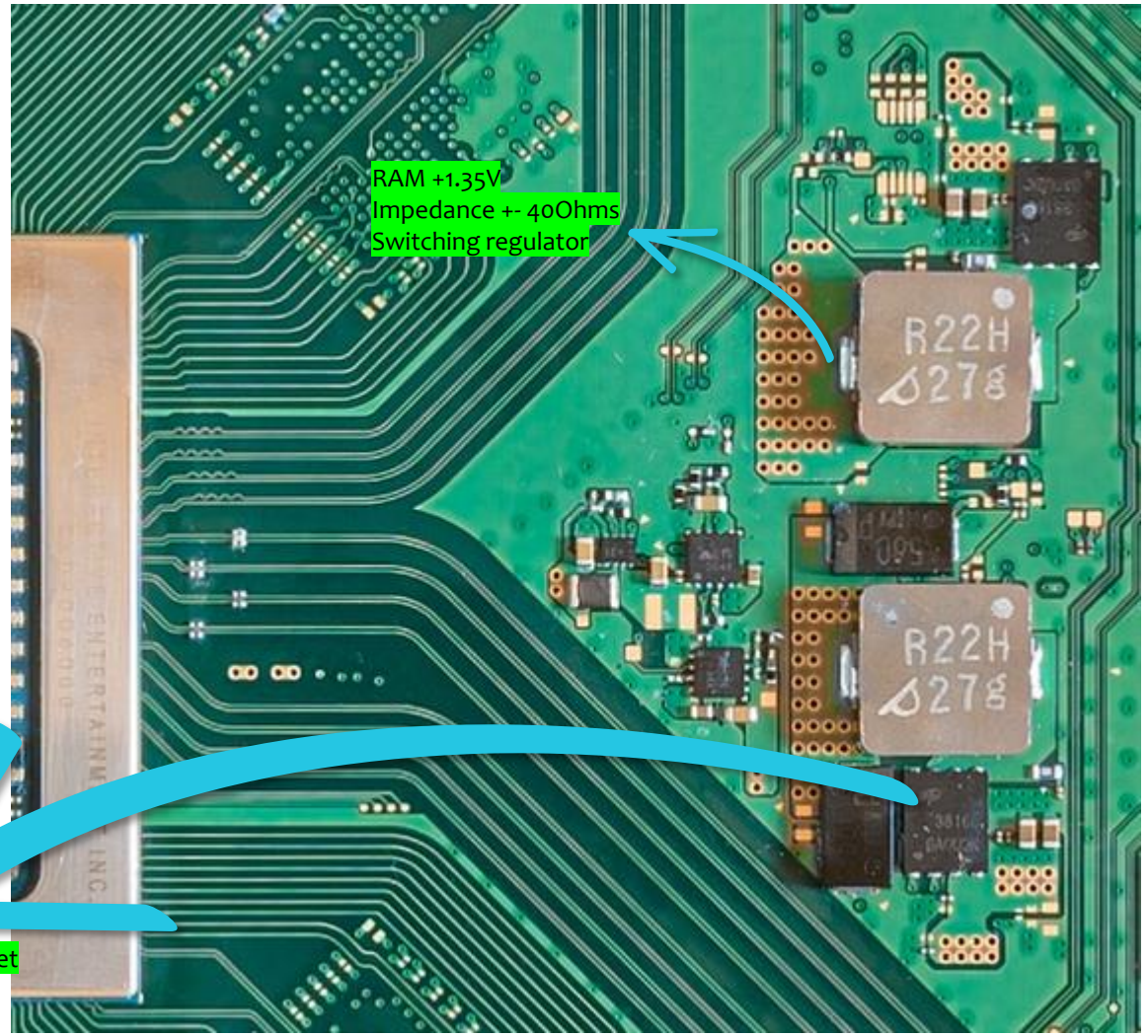
fixmyapp Research

Capacitors

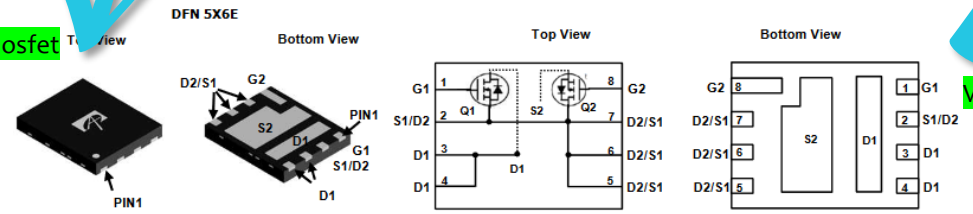

Sony Interactive Entertainment Inc.
Check capacitor for shorts
Check 12V input for shorts
Check capacitor for shorts
Check capacitor for shorts

Vmem 1.35V – Power on





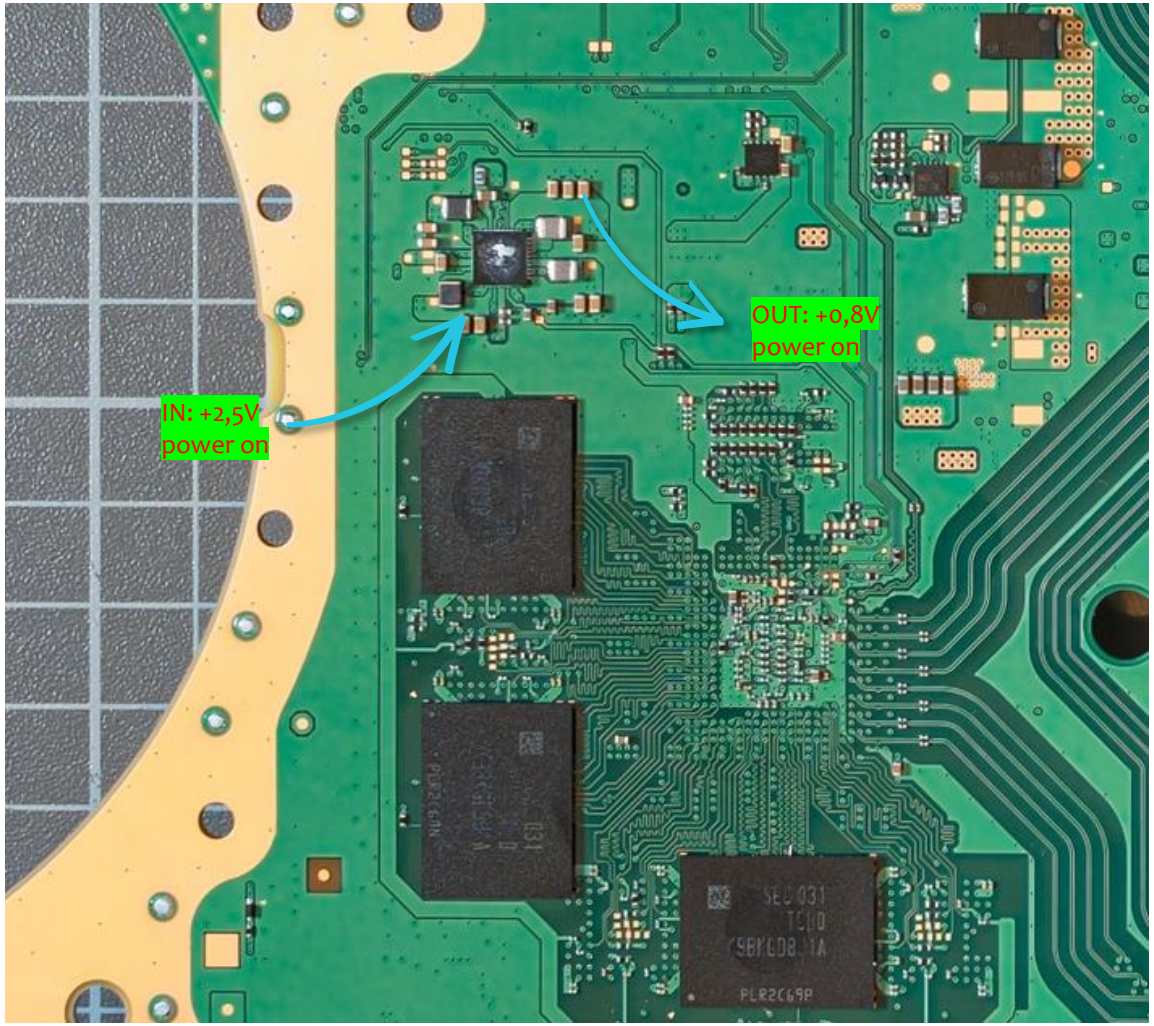
Vram Mosfet



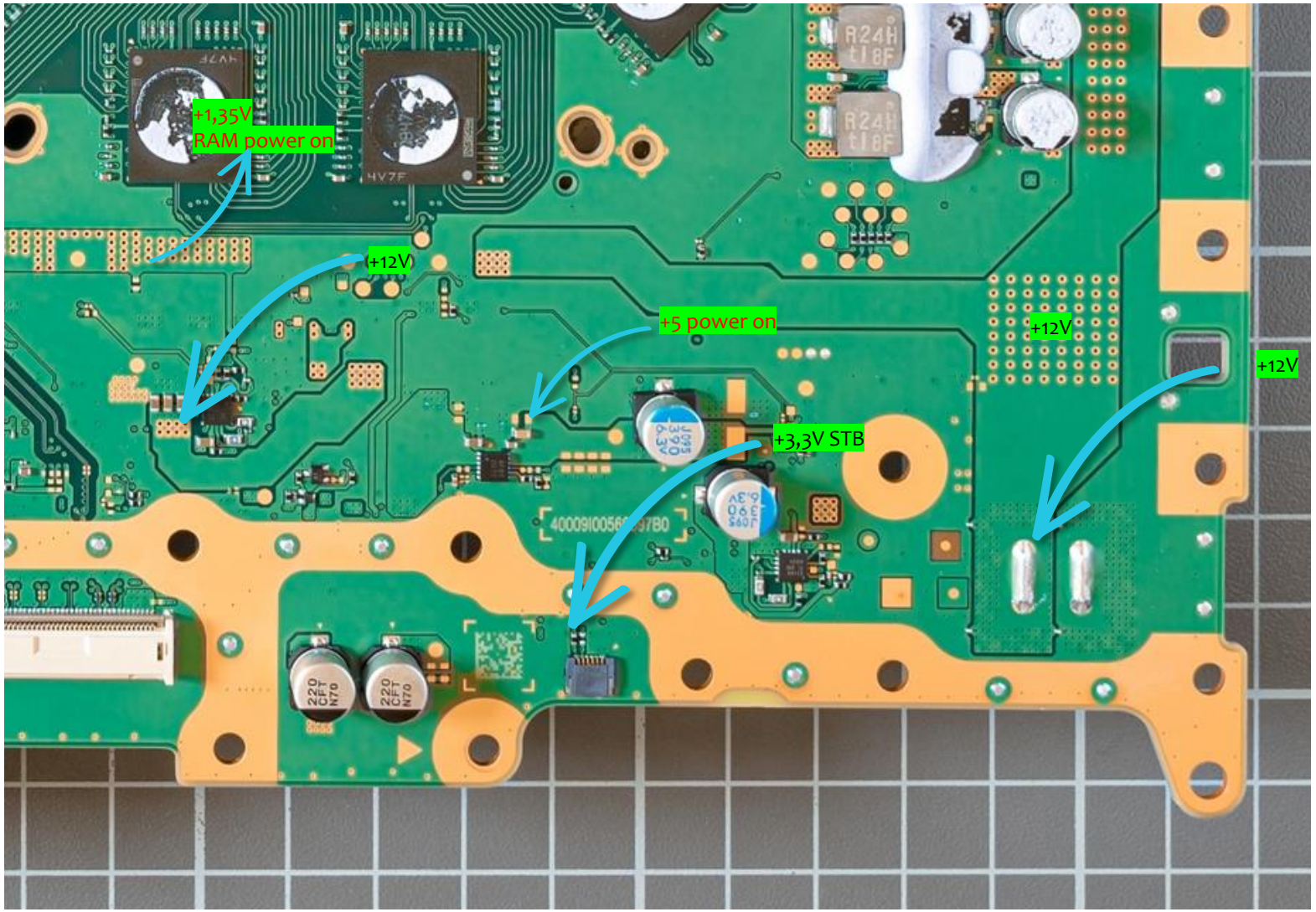
Orderable Part Number	Package Type	Form	Minimum Order Quantity
AONX38168	DFN 5x6E	Tape & Reel	3000

Vram Mosfet

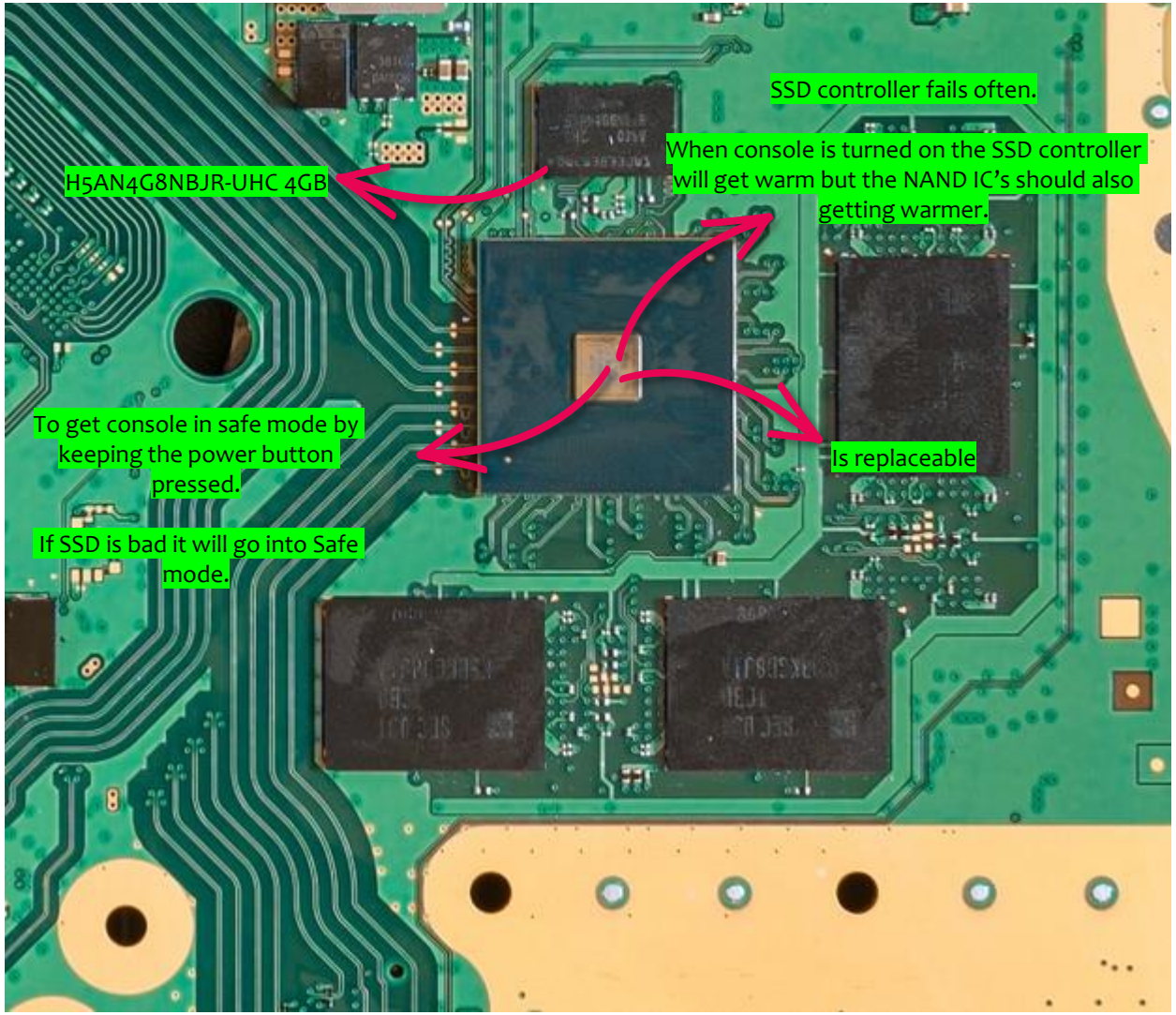
SSD voltage regulator – Power on

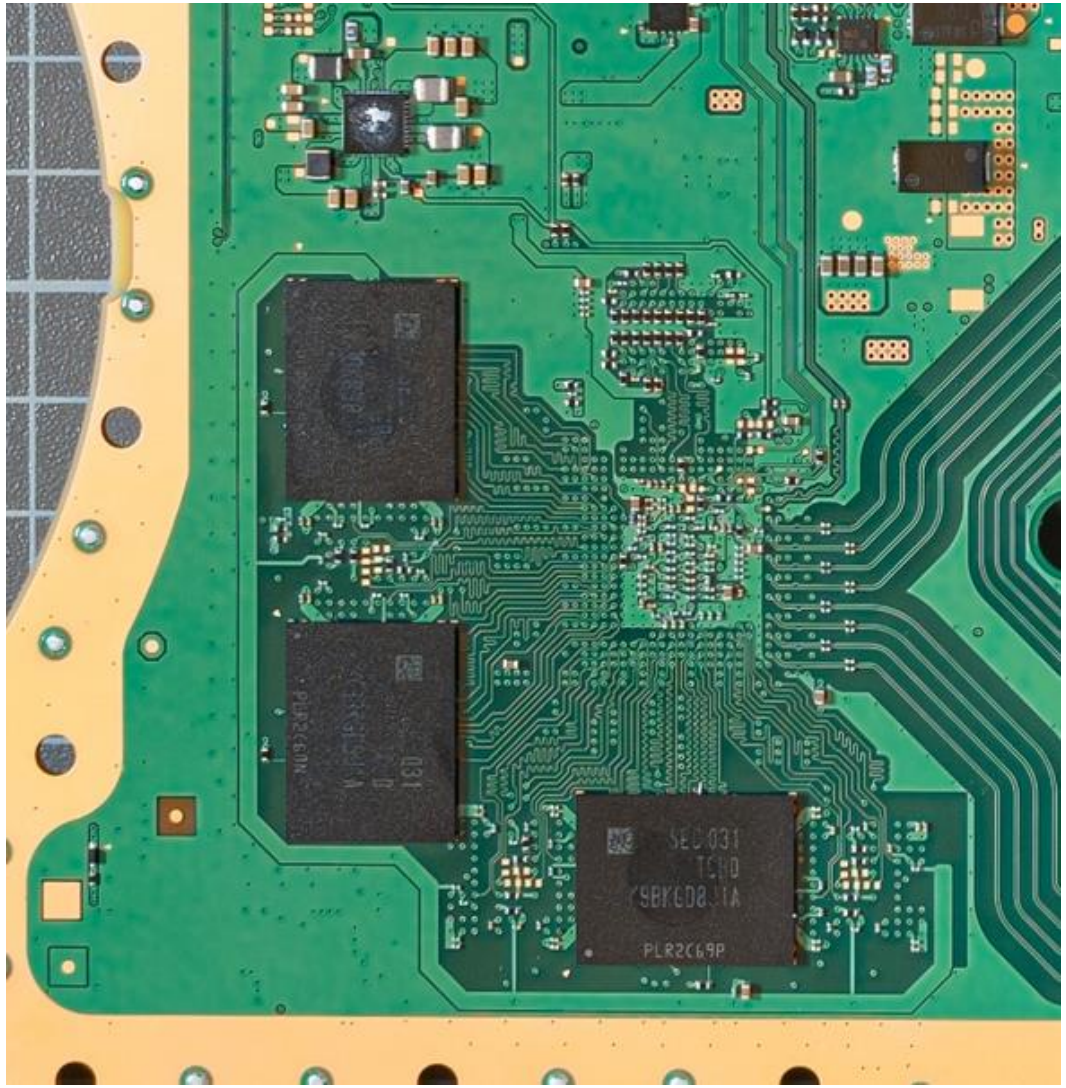


Standby and power on voltages

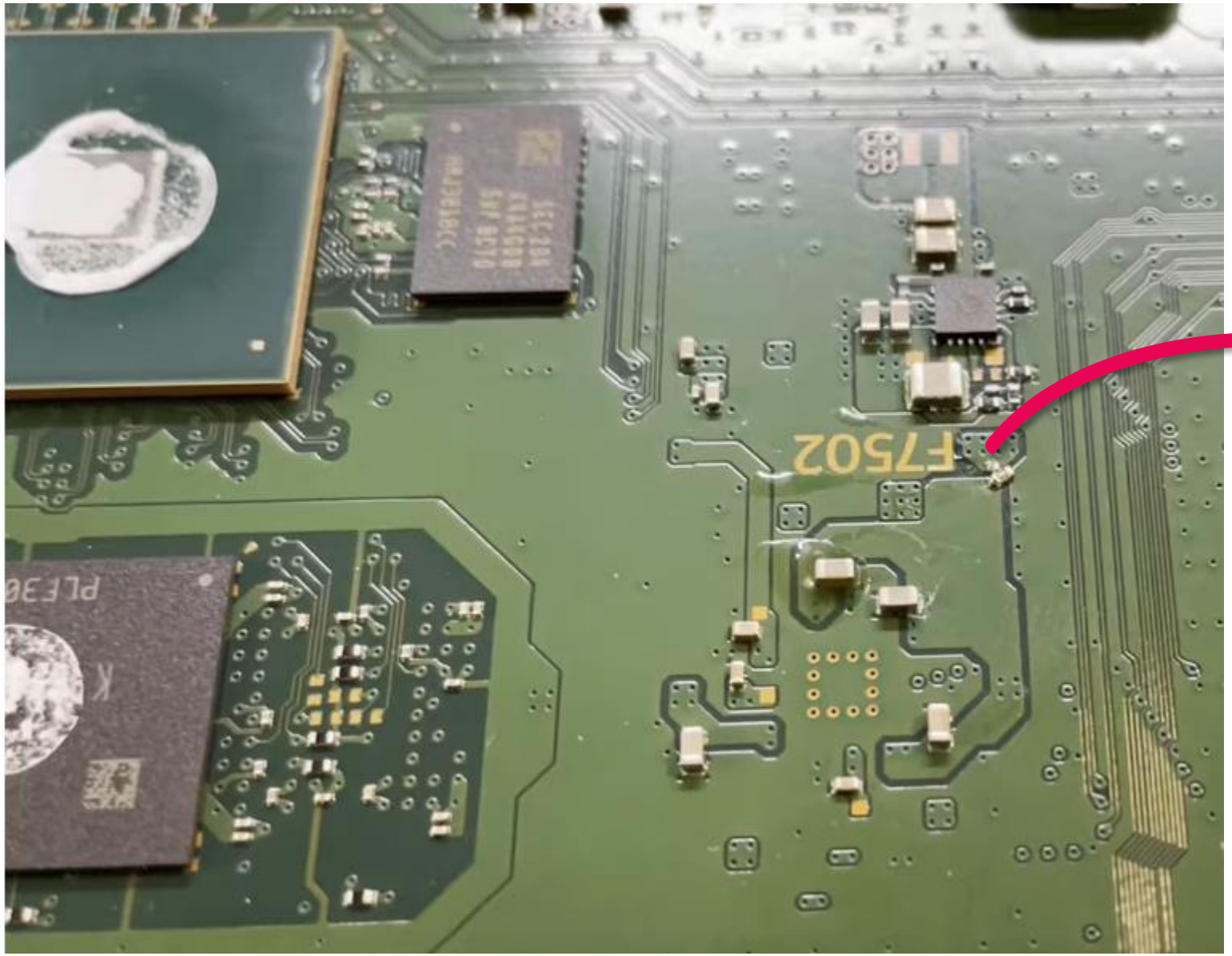


SSD-Controller & NAND flash memory





SSD controller fuse



When you see a current draw in standby of 175 mA and then back to 10Ma. There can be a bad SSD controller IC.

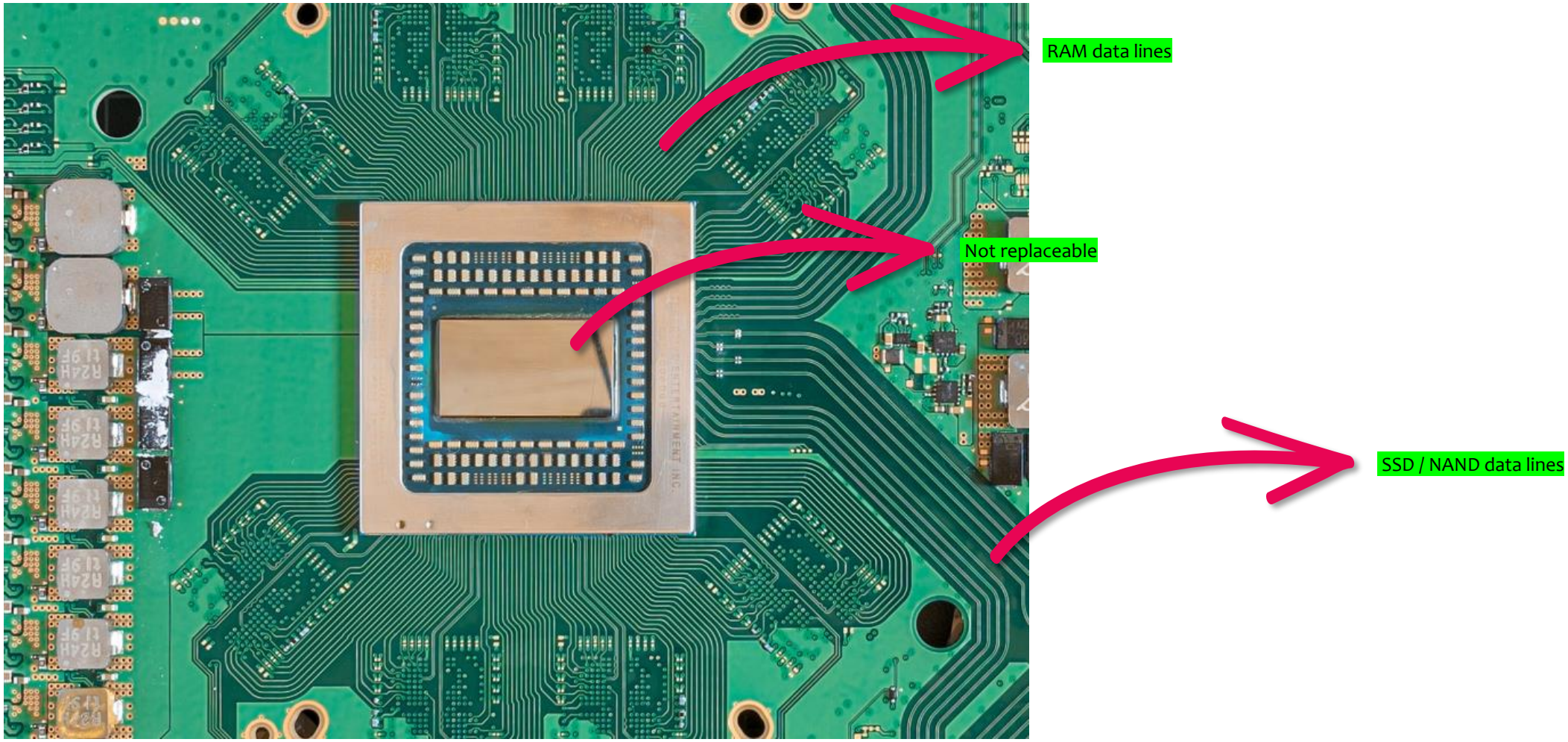
Remove fuse F7520 that feeds the voltage on the back to the voltage regulator that generates power for the SSD circuit.

Set a multimeter in current mode in series with the pads of the fuse and measure the current draw.

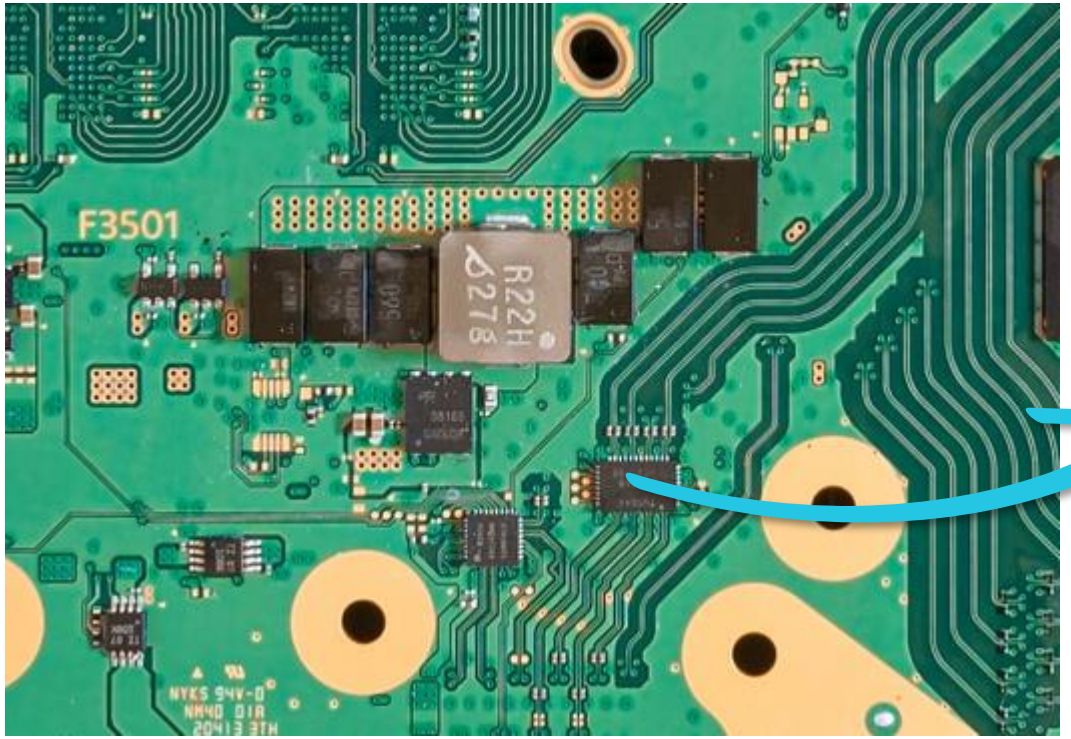
Good working SSD circuit draws: **310mA**.

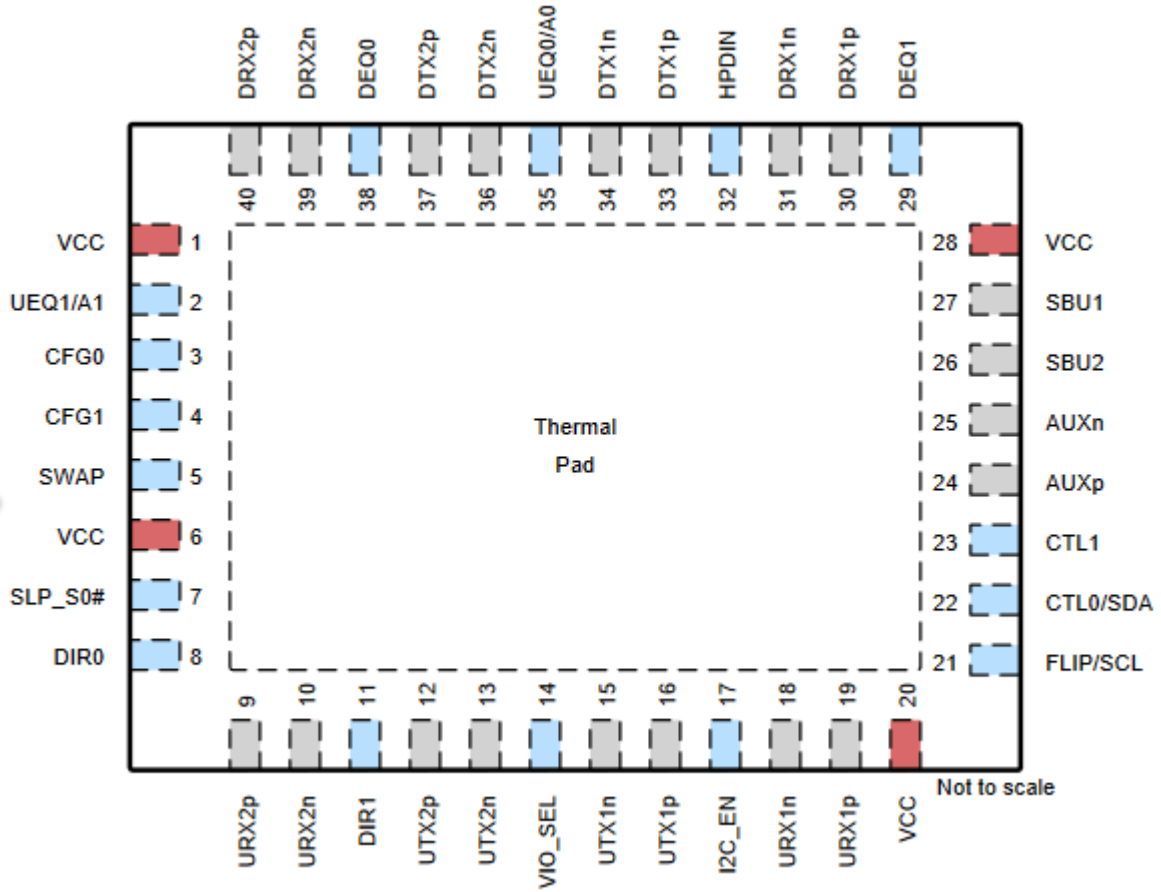
If current is different on a faulty board change the SSD controller IC.

APU

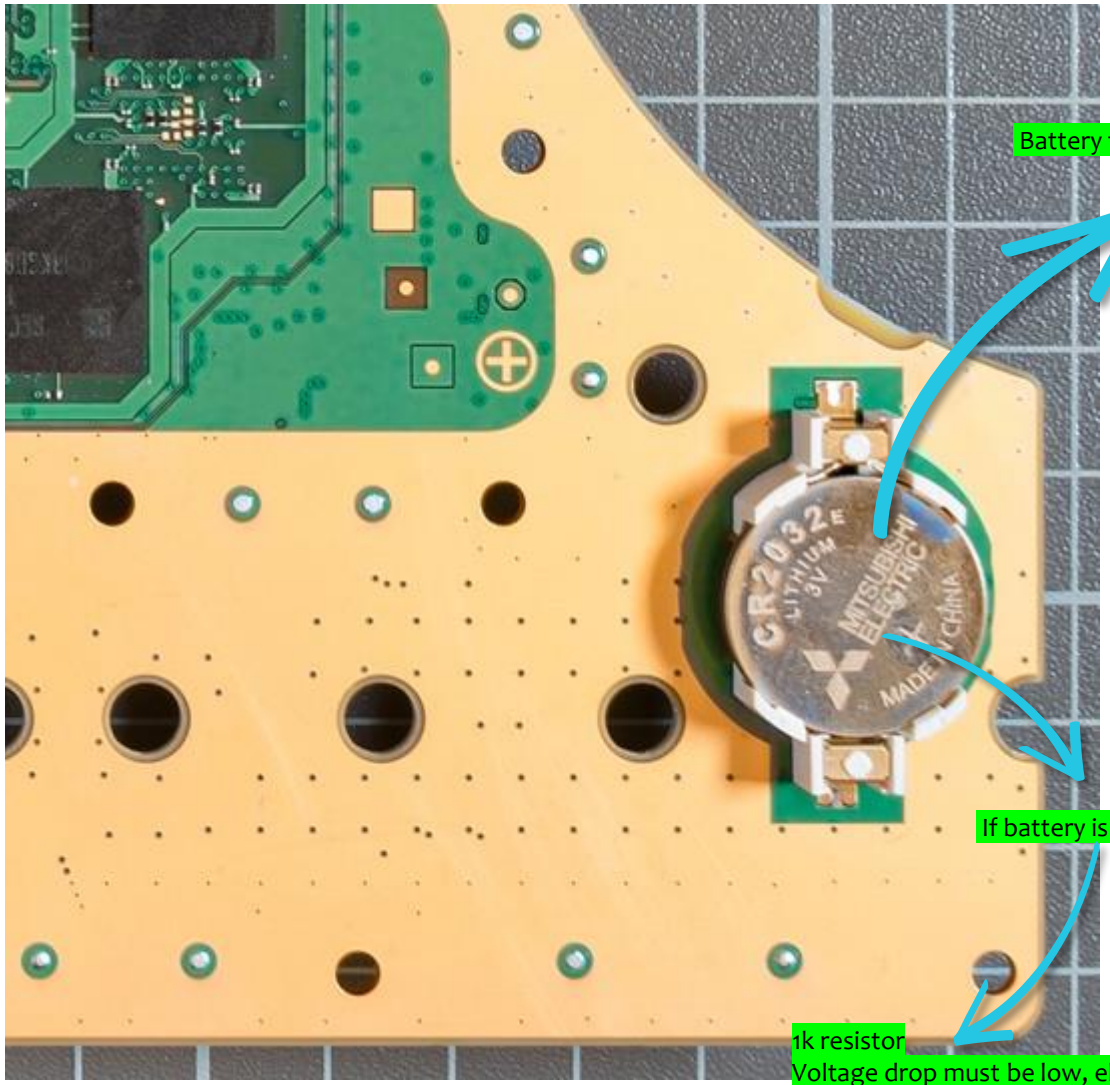


TUSB1044 - USB TYPE-C 10 Gbps Redriver





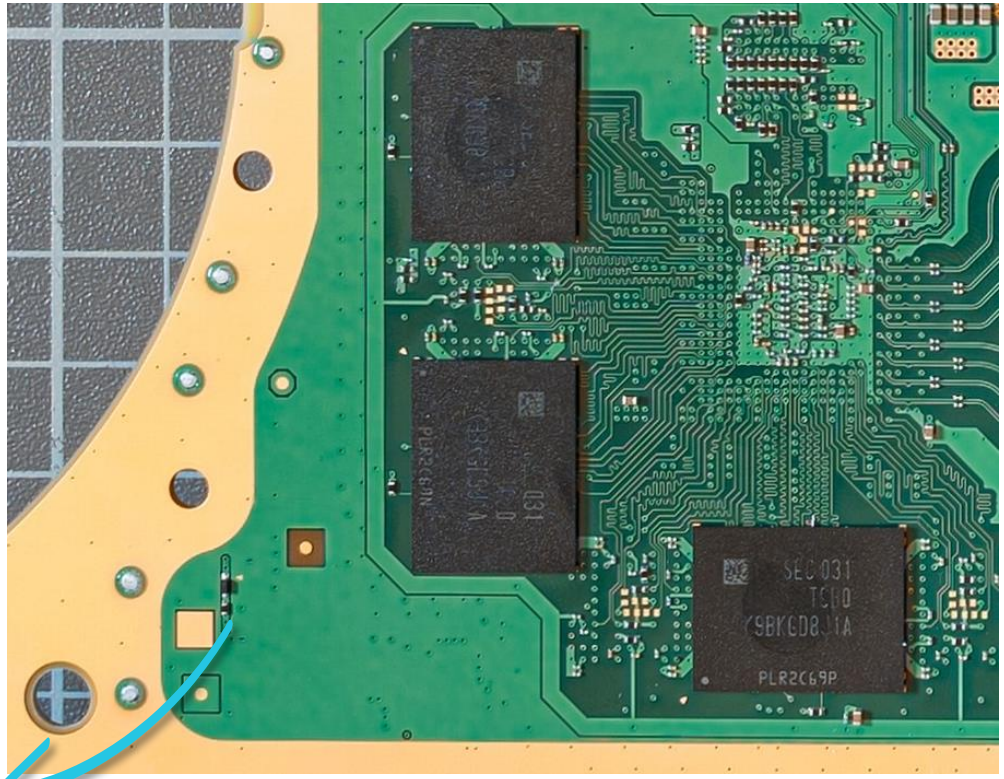
Battery +3V



Battery for real time clock it runs the 32Khz oscillator for the southbridge

If battery is empty

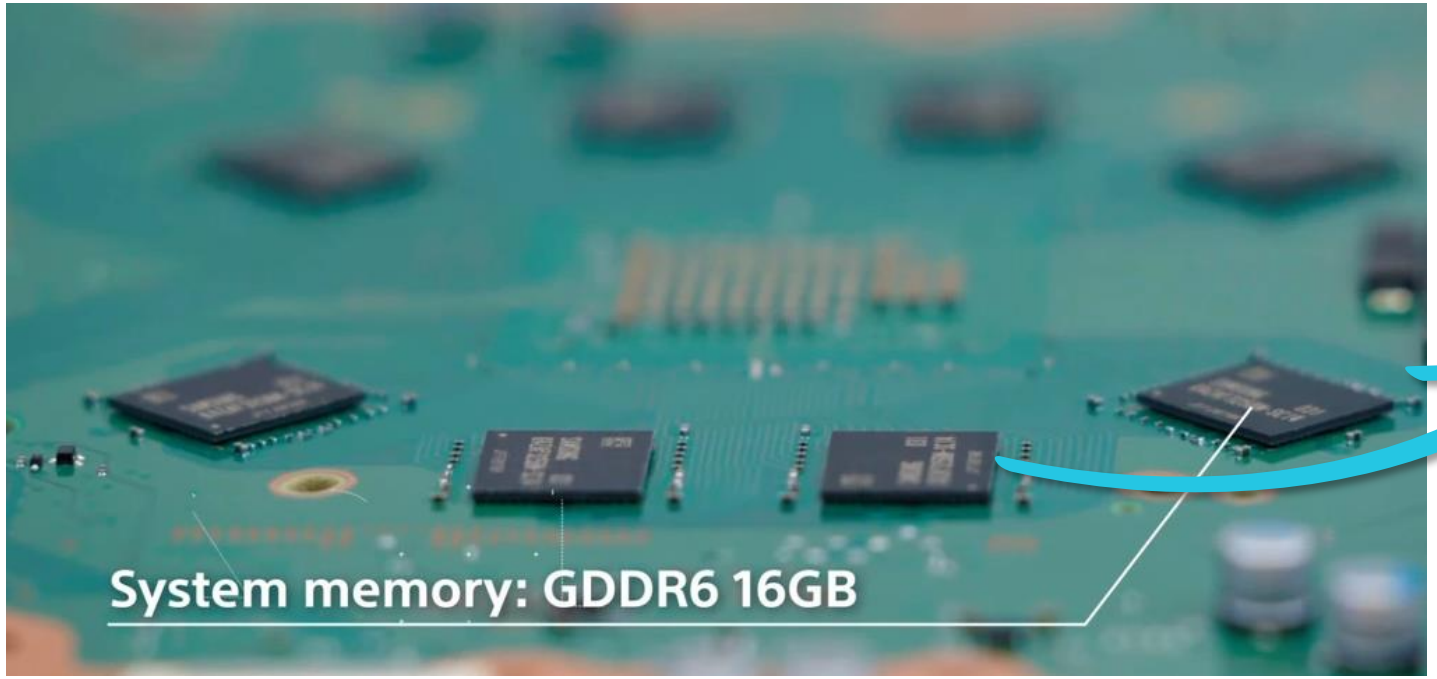
1k resistor
Voltage drop must be low, else there is a short draining the clock battery, maybe southbridge is faulty.

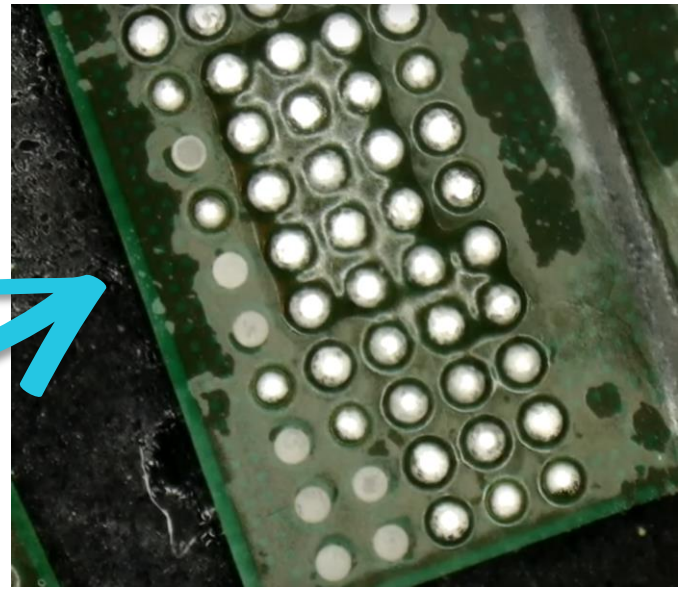


Troubleshooting RAM

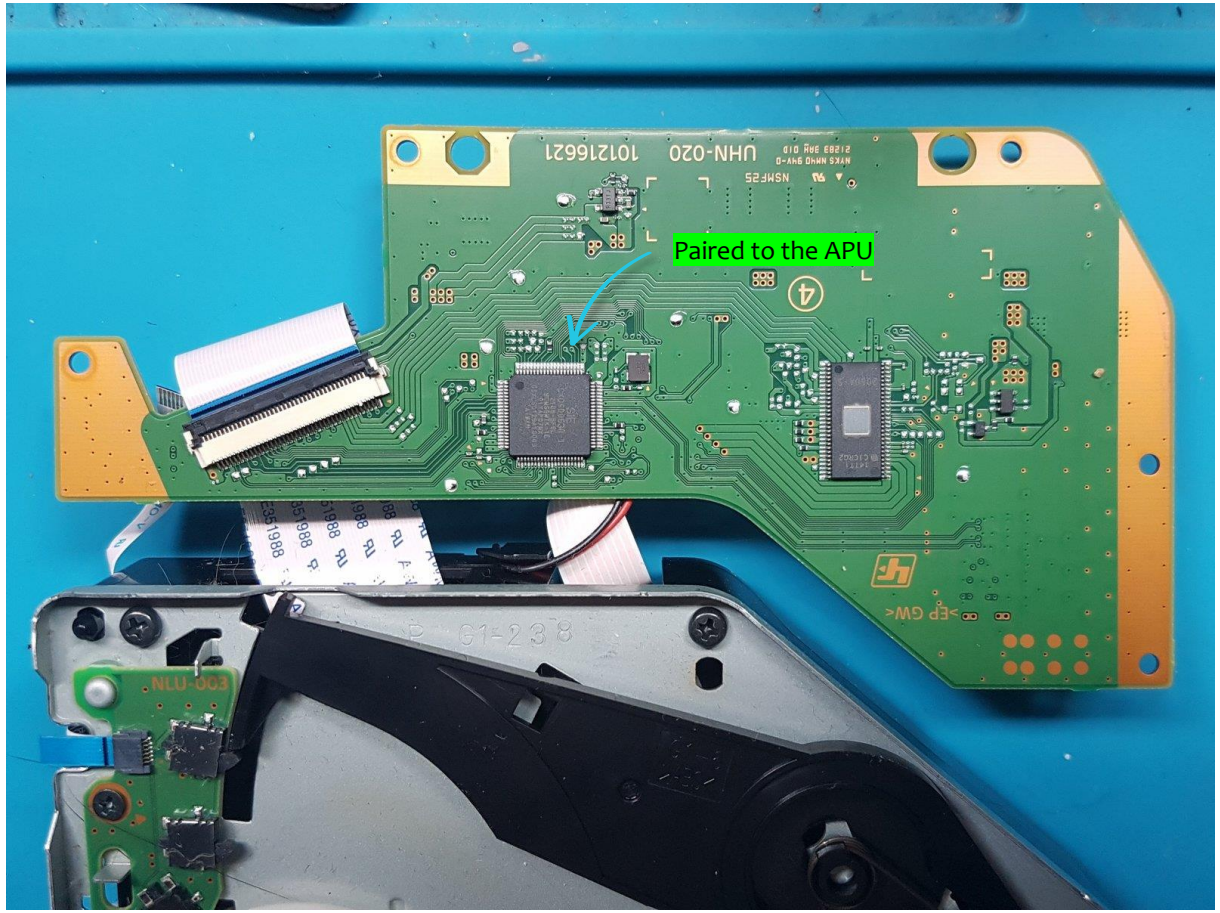
When you can’t update PS5 and during the update process or after the update it reboots then the cause can be RAM related.

Check all ram modules you can use a thermal camera to see the temperature and check if a RAM module get’s hot and reflow/replace them if problem still exist.
The SSD controller / memory modules can be faulty.





Disc drive board is paired / married to ps5 console



No online gaming after update

Can’t update console

No disc drive support

Fuses

Part Marking System

Amp Code	Marking Code
0.250	
0.375	
0.500	
0.750	
001.	
1.25	
01.5	
1.75	
002.	
02.5	
003.	
03.5	
004.	
005.	

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SUPER THANKS

